Chapter 11: Markov Chains

11.1 Markov Processes

11.1

~~11.1~~ a)
$$M_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}[X_n + (n-1)M_{n-1}]$$
$$= \frac{1}{n}X_n + (1-\frac{1}{n})M_{n-1}$$

Clearly if M_{n-1} is given then M_n depends only on X_n and is independent of $M_{n-2}, M_{n-3}, \ldots$ $\therefore$ M_n is a Markov process.

b)
$$f_{M_n}(x \mid M_{n-1} = y)dx = P[x < M_n \leq x + dx \mid M_{n-1} = y]$$
$$= P[x < \frac{1}{n} + (1-\frac{1}{n})y \leq x + dx]$$
$$= P[nx - (n-1)y < x_n \leq nx - (n-1)y + dx]$$
$$= f_x(nx - (n-1)y)dx$$

11.2

8.2 a) The number X_n of black balls in the urn completely specifies the probability of outcomes of a trial; therefore X_n is independent of its past values and X_n is a Markov process.

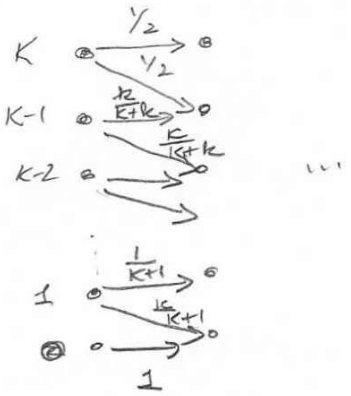
$$\begin{aligned}
P[X_n = 4|X_{n-1} = 5] &= \frac{5}{10} = 1 - P[X_n = 5|X_{n-1} = 5] \\
P[X_n = 3|X_{n-1} = 4] &= \frac{4}{9} = 1 - P[X_n = 4|X_{n-1} = 4] \\
P[X_n = 2|X_{n-1} = 3] &= \frac{3}{8} = 1 - P[X_n = 3|X_{n-1} = 3] \\
P[X_n = 1|X_{n-1} = 2] &= \frac{2}{7} = 1 - P[X_n = 2|X_{n-1} = 2] \\
P[X_n = 0|X_{n-1} = 1] &= \frac{1}{6} = 1 - P[X_n = 1|X_{n-1} = 1] \\
P[X_n = 0|X_{n-1} = 0] &= 1
\end{aligned}$$

All transition probability are independent of time.

11.2 ⓒ Let X_n = # black balls in urn

$P[X_n = k-1 \mid X_n = k] = 1 - \frac{k}{K+k}$

$P[X_n = k \mid X_n = k] = \frac{k}{K+k}$

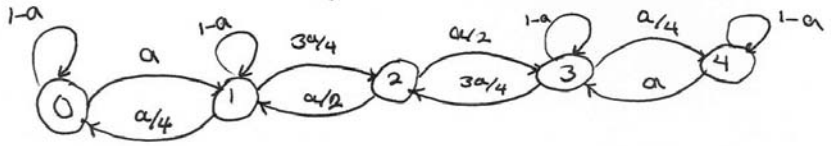


$$\begin{bmatrix}
1 & 0 & 0 & \cdots & 0 \\
\frac{K}{K+1} & \frac{1}{K+1} & 0 & \cdots & 0 \\
0 & \frac{K}{K+2} & \frac{2}{K+2} & 0 \cdots 0 & \\
 & & \ddots & & \\
 & & & \frac{1}{2} & \frac{1}{2}
\end{bmatrix}$$

11.3 a) X_n is Markov since $x_n \in \{0,1,2,3,4\} = X$ and for $\forall x_n \in X$

$$x_{n+1} = x_n \text{ or } x_{n+1} = x_n + 1 \text{ or } x_{n+1} = x_{n-1}$$

$$\Rightarrow P(X_{n+1} = x_{n+1} \mid X_n = x_n, \cdots) = P(X_{n+1} = x_{n+1} \mid X_n = x_n)$$



$$P = \begin{pmatrix} 1-a & a & 0 & 0 & 0 \\ a/4 & 1-a & \frac{3a}{4} & 0 & 0 \\ 0 & a/2 & 1-a & a/2 & 0 \\ 0 & 0 & \frac{3a}{4} & 1-a & a/4 \\ 0 & 0 & 0 & a & 1-a \end{pmatrix}$$

(b) Transition probabilities are independent from n as P shows.

(c) If $a=1$ then in all events a change in the color of the selected ball takes place and there is no self-loop. The transition matrix would be:

$$P_2 \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(d) $P(X_{n+1} = m+1 \mid X_n = m) = \underbrace{P(m+1 \mid m)}_{\text{simplified notation}} = \frac{m}{K} a \qquad \forall\, 1 < m < K,\ m \in N$

$$P(X_{n+1} = m-1 \mid X_n = m) = P(m-1 \mid m) = \frac{K-m}{m} a \qquad \forall\, 1 < m < K,\ m \in N$$

$P(m \mid m) = 1-a$

$P(m+1 \mid m) = a \quad m=1$

$P(m-1 \mid m) = a \quad m=K$

obviously X_n is Markov.

(11.3/d) (cont.)

$$P = \begin{pmatrix} 1-a & 0 & 0 & \cdots & 0 \\ a/K & 1-a & \frac{K-1}{K}a & \cdots & 0 \\ 0 & 2a/K & 1-a & \frac{K-2}{K}a & \cdots 0 \\ \vdots & & \frac{3a}{K} & \ddots & \ddots \\ & & & a & 1-a \end{pmatrix}$$

independent of n

in case of $a=1$, there is always a change in color:

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ \frac{1}{K} & 0 & \frac{K-1}{K} & 0 & \cdots & 0 \\ 0 & \frac{2}{K} & 0 & \frac{K-2}{K} & \cdots & 0 \\ \vdots & & & & & \\ & & & & 1 & 0 \end{pmatrix}$$

11.4 (a) X_n is Markov since $\begin{cases} X_{n+1} = X_n \pm 1 & \text{for } X_n \in \{1,2,3\} \\ X_{n+1} = X_n + 1 & \text{if } X_n = 0 \\ X_{n+1} = X_n - 1 & \text{if } X_n = 4 \end{cases}$

we show:

$P(X_{n+1} = a \mid X_n = b) = P(a|b)$

$P(1|0) = 1$

$P(0|1) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

$P(1|1) = \frac{1}{4} \times \frac{3}{4} + \frac{3}{4} \times \frac{1}{4} = \frac{6}{16}$

$P(1|2) = \frac{4}{16}$

$P(2|2) = \frac{2}{4} \times \frac{2}{4} + \frac{2}{4} \times \frac{2}{4} = \frac{8}{16}$

$P(3|2) = \frac{2}{4} \times \frac{2}{4} = \frac{4}{16}$

$P(3|3) = \frac{3}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{3}{4} = \frac{6}{16}$

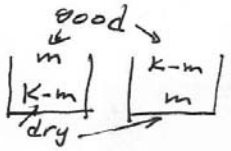
$P(4|3) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

$P(2|3) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{1}{16} & \frac{6}{16} & \frac{9}{16} & 0 & 0 \\ 0 & \frac{4}{16} & \frac{8}{16} & \frac{4}{16} & 0 \\ 0 & 0 & \frac{9}{16} & \frac{6}{16} & \frac{1}{16} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

11.4 (b) Transition probabilities do not depend on n.

(c) $P(X_{n+1} = a \mid X_n = b) = P(a|b)$ ← notation



$\forall\ 1 \le m < K$

$P(m|m) = \frac{m}{K} \times \frac{K-m}{K} + \frac{K-m}{K} \times \frac{m}{K} = \frac{2m(K-m)}{K^2}$

$P(m-1|m) = \frac{m}{K} \times \frac{m}{K} = \frac{m^2}{K^2}$

$P(m+1|m) = 1 - \frac{m^2}{K^2} - \frac{2m(K-m)}{K^2} = \frac{(K-m)^2}{K^2}$

$P(1|0) = 1$

$P(K-1|K) = 1$

11.5

8.3 If $X(t)$ has independent increments, then

$$f_{X(t_{k+1})X(t_k)\ldots X(t_1)}(x_{k+1},\ldots,x_1) = f_{X(t_1)}(x_1)f_{X(t_2)-X(t_1)}(x_2-x_1)\ldots f_{X(t_{k+1})-X(t_k)}(x_{k+1}-x_k)$$

$$\begin{aligned}
\Rightarrow f_{X(t_{k+1})}(x_k|x(t_k) &= x_k,\ldots,x(t_1)=x_1) \\
&= \frac{f_{X(t_{k+1})\ldots x(t_1)}(x_{k+1},\ldots,x_1)}{f_{X(t_k)\ldots x(t_1)}(x_k,\ldots,x_1)} \\
&= \frac{f_{X(t_1)}(x_1)\ldots f_{X(t_k)-x(t_{k-1})}(x_k-x_{k-1})f_{X(t_{k+1})-x(t_k)}(x_{k+1}-x_k)}{f_{X(t_1)}(x_1)\ldots f_{X(t_k)-x(t_{k-1})}(x_k-x_{k-1})} \\
&= f_{X(t_{k+1})-x(t_k)}(x_{k+1}-x_k) \\
&= f_{X(t_{k+1})}(x_{k+1}|x(t_k)=x_k)
\end{aligned}$$

$\Rightarrow X(t)$ is a Markov process.

However if $X(t)$ is Markov, then $X(t)$ need not have independent increments. To see why consider the Markov chain in Problem 8.2. Since X_n is Markov we have:

$$P[X_i = i, X_2 = j] = P[X_1 = i]P[X_2 = j|X_1 = i]$$

If X_n had independent increments, we would have

$$P[X_1 = i, X_2 = j] = P[X_1 = i]P[X_2 - X_1 = j - i]$$

so we require that

$$P[X_2 = j|X_1 = i] = P[X_2 - X_1 = j - i] .$$

Let $j_1 = 5$, $i_1 = 4$ then

$$P[X_2 = 4|X_1 = 5] = \frac{1}{2} \stackrel{?}{=} P[X_2 - X_1 = -1]$$

but if $j_2 = 4$, $i_2 = 3$ then

$$P[X_2 = 3|X_1 = 4] = \frac{4}{9} \stackrel{?}{=} P[X_2 - X_1 = -1] .$$

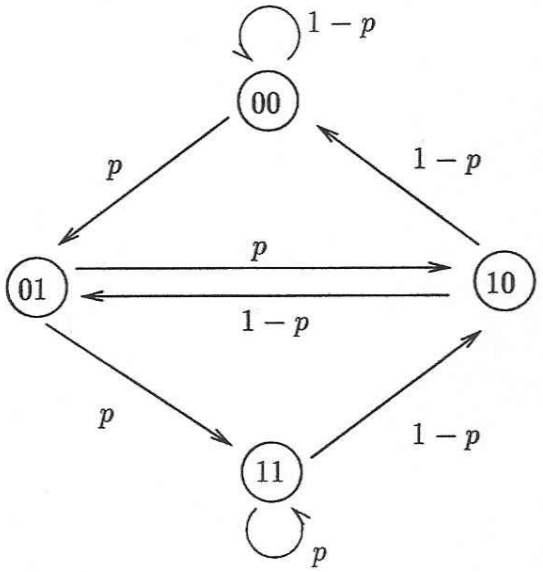
Thus X_n cannot have independent increments.

11.6

$\underline{Z}_n = (x_n, x_{n-1})$

$$\begin{aligned} P[\underline{Z}_{n+1} &= (x_{n+1}, x_n) | \underbrace{\underline{Z}_n = (x_n, x_{n-1}), \underline{Z}_{n-1} = (x_{n-1}, x_{n-2}) \ldots}_{\text{all past vectors}}] \\ &= P[\underline{Z}_{n+1} = (x_{n+1}, x_n) | \underbrace{X_n = x_n, X_{n-1} \ldots}_{\text{all past Bernoulli trials}}] \\ &= P[\overbrace{X_{n+1} = x_{n+1}}^{\text{next trial}}] \\ &= P[\underline{Z}_{n+1} = (x_{n+1}, x_n) | \underline{Z}_n = (x_n, x_{n-1})] \end{aligned}$$

$\therefore \underline{Z}_n$ is a Markov process.



where $p = P[x = 1]$

11.7

$Y_n = rY_{n-1} + X_n \qquad Y_0 = 0 \qquad Y_n - rY_{n-1} = X_n$

$$f_{Y_n}(y|y_{n-1} = y_1 \;\; Y_{n-2} = y_2 \ldots) = f_{X_n}(y - ry_1)$$
$$= f_{Y_n}(y|Y_{n-1} = y_1)$$

$\Rightarrow Y_n$ is a Markov process.

(b)

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-m)^2/2\sigma^2}$$

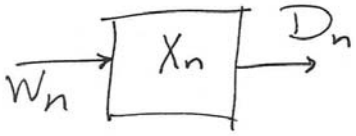
$$f_{Y_{n+1}}(y \mid Y_n = y_n)\,dy = P[y < Y_{n+1} \le y + dy \mid Y_n = y_n]$$

$$= P[y_{n+1} < ry_n + X_{n+1} \le y_{n+1} + dy]$$

$$= P[y_{n+1} - ry_n < X_{n+1} \le y_{n+1} - ry_n + dy]$$

$$= f_{X_{n+1}}(y_{n+1} - ry_n)\,dy$$

11.8



(a)

$X_2 = X_1 + (W_1 - D_1)$
$X_3 = X_2 + (W_2 - D_2)$
$\vdots$
$X_{n+1} = X_n + (W_n - D_n)$

(b) According to the above recursive equations, X_{n+1} depends only on X_n plus $(W_n - D_n)$. The X_n is Markov only if W_n and D_n are independent random variables.

11.2 Discrete-Time Markov Chains

11.9

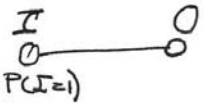
~~8.7~~ Assume X_n is discrete, then

$$\begin{aligned} P[X_n = x | X_{n-1} = x_1, X_{n-2} = x_2, ...] &= P[X_n = x] \quad \text{since } X_n \text{ is iid} \\ &= P[X_n = x | X_{n-1} = x_1] \end{aligned}$$

$\therefore X_n$ is a Markov process with transition probabilities:

$$P[X_n = x | X_{n-1} = x_1] = P[X_n = x] \quad \text{all } x_1$$

11.10 (a)



$$P(O=0) = P(O=0|I=0)P(I=0) + P(O=0|I=1)P(I=1)$$
$$= (1-\varepsilon)a + \varepsilon(1-a) = a + \varepsilon - 2a\varepsilon$$

$$P(O=1) = P(O=1|I=0)P(I=0) + P(O=1|I=1)P(I=1)$$
$$= \varepsilon a + (1-\varepsilon)(1-a) = 1 - a - \varepsilon + 2a\varepsilon$$

(b) $P = \begin{pmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{pmatrix}$ in a system of K independent symmetric channels:

$$P(O_{n+1}=o_{n+1} | O_n=o_n, \ldots) = P(O_{n+1}=o_{n+1} | O_n=o_n) \quad \text{Markov}$$

since if we know O_n, the O_{n+1} could be found (channels are independent)

(c) $$P_k = (P)^k = \begin{pmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{pmatrix}^k \Rightarrow \begin{aligned} P_k(0) &= (1-\varepsilon)P_{k-1}(0) + \varepsilon P_{k-1}(1) \\ P_k(1) &= \varepsilon P_{k-1}(0) + (1-\varepsilon)P_{k-1}(1) \end{aligned}$$

$(P_k(0), P_k(1))^t$ is the output probability of channel k

To find a close form for P^k we can use diagonalization method

$|\lambda I - P| = 0 \Rightarrow (\lambda + \varepsilon - 1)^2 - \varepsilon^2 = 0 \Rightarrow \begin{cases} \lambda = 1 \\ \lambda = 1 - 2\varepsilon \end{cases}$

$\lambda = 1 \Rightarrow v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $\lambda = 1-2\varepsilon \Rightarrow v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ $U = (v_1 \; v_2) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$P = UDU^{-1}$ $D = \begin{pmatrix} 1 & 0 \\ 0 & 1-2\varepsilon \end{pmatrix}$

$\Rightarrow$

$$P = UDU^{-1} = -\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1-2\varepsilon \end{pmatrix}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$P^k = (UDU^{-1})^k = UD^kU^{-1} = -\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & (1-2\varepsilon)^k \end{pmatrix}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\Rightarrow P^k = -\frac{1}{2}\begin{pmatrix} -1-(1-2\varepsilon)^n & -1+(1-2\varepsilon)^n \\ -1+(1-2\varepsilon)^n & -1-(1-2\varepsilon)^n \end{pmatrix}$$

$$P^k = \begin{pmatrix} \frac{1}{2}+\frac{1}{2}(1-2\varepsilon)^n & \frac{1}{2}+\frac{1}{2}(1-2\varepsilon)^n \\ \frac{1}{2}-\frac{1}{2}(1-2\varepsilon)^n & \frac{1}{2}+(1-2\varepsilon)^n \end{pmatrix}$$

If $k \to \infty$ for any $|1-2\varepsilon| < 1$ $(1-2\varepsilon)^k \to 0$

$$\Rightarrow P^\infty \to \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

and the steady state probability vector is $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$

(d) When $k \to \infty$ $\begin{cases} P_k(0) = P_{k-1}(0) \\ P_k(1) = P_{k-1}(1) \end{cases} \Rightarrow \begin{cases} P_k(0) = (1-\varepsilon)P_k(0) + \varepsilon P_k(1) \\ P_k(1) = \varepsilon P_k(0) + (1-\varepsilon)P_k(1) \end{cases}$

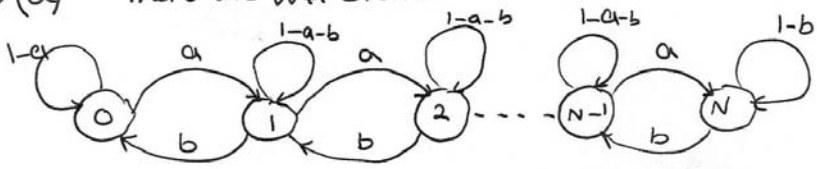
$\Rightarrow P_k(0) = P_k(1)$ } $\Rightarrow$ $P_k(0) = P_k(1) = 0.5$ output $\underset{k\to\infty}{=} \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$

we know $P_k(0) + P_k(1) = 1$

11.11

Version 1: Arrival and Departure are mutually exclusive.

(a) There are $N+1$ states



Clearly every state x_{n+1} be found either from x_n or $x_n \pm 1 \Rightarrow$ Markov

(b)

$$P_{(N+1)\times(N+1)} = \begin{pmatrix} 1-a & a & 0 & 0 & \cdots & 0 \\ b & 1-a-b & a & 0 & \cdots & 0 \\ 0 & b & 1-a-b & a & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & b & 1-a-b & a \\ 0 & 0 & \cdots & 0 & 1-b & b \end{pmatrix}$$

(c) $xP = x \Rightarrow (1-a)x_0 + bx_1 = x_0 \Rightarrow x_1 = \frac{a}{b}x_0$

$ax_0 + (1-a-b)x_1 + bx_2 = x_1 \Rightarrow (a+b)\frac{a}{b}x_0 - ax_0 = bx_2 \Rightarrow x_2 = \frac{a^2}{b^2}x_0$

We claim $x_k = \frac{a^k}{b^k}x_0$

for $k = 1, 2$ it's proved. Now if it is true for $k-1, k$ we show it's valid for $k+1$ as well:

$$a x_{k-1} + (1-a-b)x_k + b x_{k+1} = x_k$$

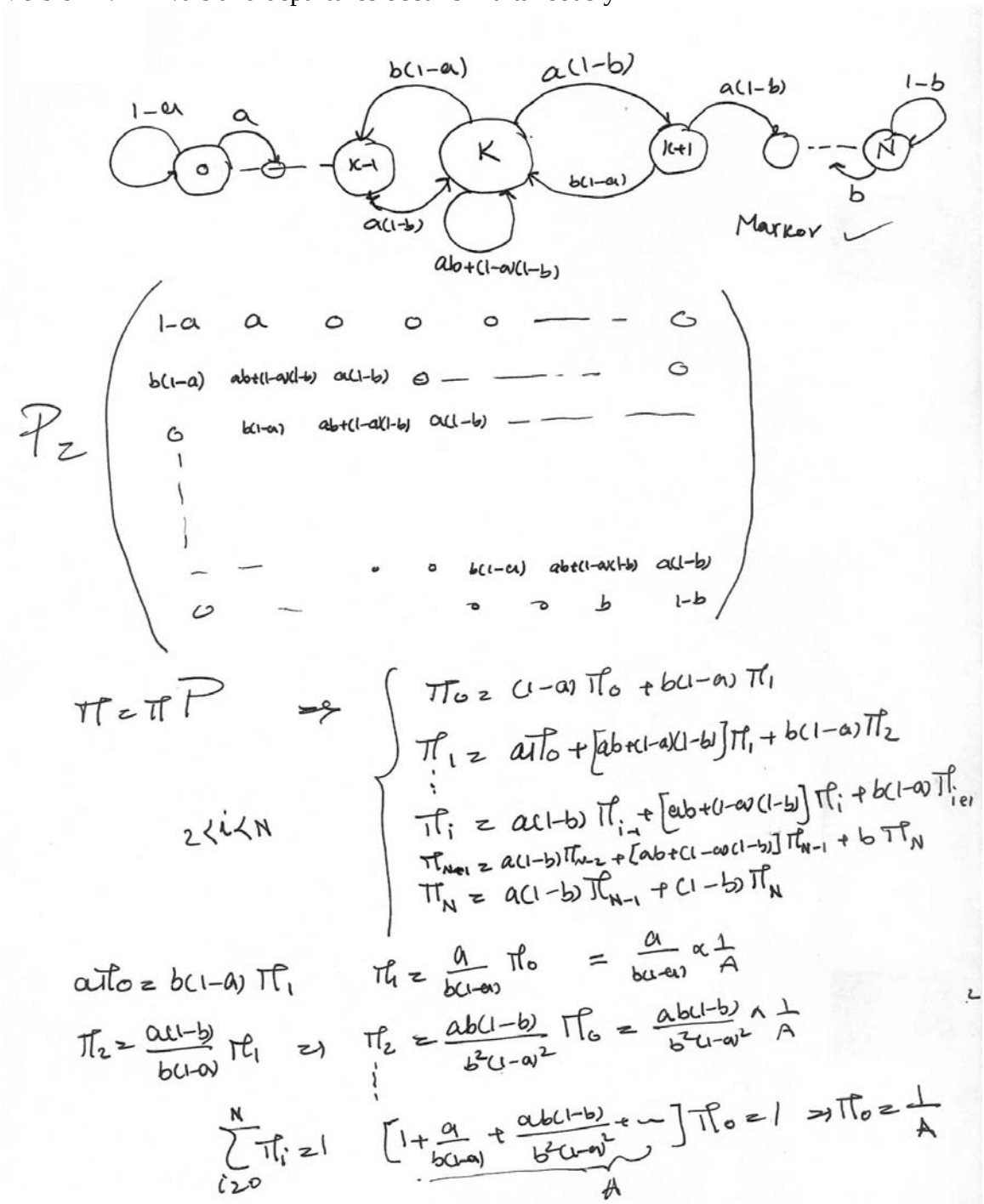
$$\Rightarrow a \times \frac{a^{k-1}}{b^{k-1}}x_0 + b\,x_{k+1} = (a+b)\frac{a^k}{b^k}x_0 \Rightarrow b\,x_{k+1} = \frac{a^k}{b^{k-1}}\left(1+\frac{a}{b}-1\right)x_0$$

$$\Rightarrow x_{k+1} = \frac{a^{k+1}}{b^{k+1}}x_0$$

Finally $x_0 + x_1 + \cdots + x_N = 1 \Rightarrow x_0\left(1 + \frac{a}{b} + \frac{a^2}{b^2} + \cdots + \frac{a^N}{b^N}\right) = 1$

$$\Rightarrow x_0 = \frac{1}{\sum_{i=0}^{N}\left(\frac{a}{b}\right)^i} \Rightarrow \boxed{x = (x_0, x_1, \cdots, x_N), \quad x_i = \frac{\left(\frac{a}{b}\right)^i}{\sum_{j=0}^{N}\left(\frac{a}{b}\right)^j}}$$

Version 2: Arrivals and departures occur simultaneously



11.12 **a)**
$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & \frac{5}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{7} & \frac{5}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{8} & \frac{5}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{4}{9} & \frac{5}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{5}{10} & \frac{5}{10} \end{bmatrix}$$

b)
$$\mathbf{P}^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{11}{36} & \frac{25}{36} & 0 & 0 & 0 & 0 \\ \frac{1}{21} & \frac{65}{144} & \frac{25}{49} & 0 & 0 & 0 \\ 0 & \frac{3}{28} & \frac{225}{448} & \frac{25}{64} & 0 & 0 \\ 0 & 0 & \frac{1}{6} & \frac{95}{192} & \frac{25}{81} & 0 \\ 0 & 0 & 0 & \frac{2}{9} & \frac{19}{36} & \frac{1}{4} \end{bmatrix}$$

$$p_{54}(2) = \frac{19}{36} \quad \text{from } \mathbf{P}^2$$

$$p_{54}(2) = p_{55}^{\text{no change}}(1)p_{54}^{\text{change}}(1) + p_{54}^{\text{change}}(1)p_{44}^{\text{no change}}(1)$$

$$= \frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{5}{9} = \frac{19}{36} \quad \surd$$

c) As $n \to \infty$ eventually all black balls are removed. Thus

$$\mathbf{P}^n \to \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

11.13 (a) $a = \frac{1}{10} \Rightarrow$

$$P = \begin{pmatrix} 0.9 & 0.1 & 0 & 0 & 0 \\ 0.025 & 0.9 & 0.075 & 0 & 0 \\ 0 & 0.05 & 0.9 & 0.05 & 0 \\ 0 & 0 & 0.075 & 0.9 & 0.025 \\ 0 & 0 & 0 & 0.1 & 0.9 \end{pmatrix}$$

(b)

11.13.b

P =

0.9000	0.1000	0	0	0
0.0250	0.9000	0.0750	0	0
0	0.0500	0.9000	0.0500	0
0	0	0.0750	0.9000	0.0250
0	0	0	0.1000	0.9000

P^2 =

0.8125	0.1800	0.0075	0	0
0.0450	0.8163	0.1350	0.0037	0
0.0013	0.0900	0.8175	0.0900	0.0013
0	0.0037	0.1350	0.8163	0.0450
0	0	0.0075	0.1800	0.8125

P^4 =

0.6683	0.2939	0.0365	0.0014	0.0000
0.0735	0.6865	0.2214	0.0183	0.0003
0.0061	0.1476	0.6926	0.1476	0.0061
0.0003	0.0183	0.2214	0.6865	0.0735
0.0000	0.0014	0.0365	0.2939	0.6683

P^8 =

0.4684	0.4035	0.1151	0.0126	0.0004
0.1009	0.5259	0.3121	0.0580	0.0031
0.0192	0.2081	0.5455	0.2081	0.0192
0.0031	0.0580	0.3121	0.5259	0.1009
0.0004	0.0126	0.1151	0.4035	0.4684

P^∞ =

0.0625	0.2500	0.3750	0.2500	0.0625
0.0625	0.2500	0.3750	0.2500	0.0625
0.0625	0.2500	0.3750	0.2500	0.0625
0.0625	0.2500	0.3750	0.2500	0.0625
0.0625	0.2500	0.3750	0.2500	0.0625

(c)

11.13.c

P =

0	1.0000	0	0	0
0.2500	0	0.7500	0	0
0	0.5000	0	0.5000	0
0	0	0.7500	0	0.2500
0	0	0	1.0000	0

P^2

0.2500	0	0.7500	0	0
0	0.6250	0	0.3750	0
0.1250	0	0.7500	0	0.1250
0	0.3750	0	0.6250	0
0	0	0.7500	0	0.2500

P^4 =

0.1563	0	0.7500	0	0.0938
0	0.5313	0	0.4688	0
0.1250	0	0.7500	0	0.1250
0	0.4688	0	0.5313	0
0.0938	0	0.7500	0	0.1563

P^{50} =

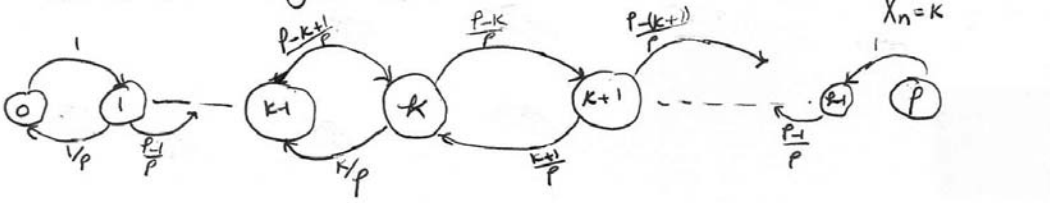
0.1270	0	0.7500	0	0.1230
0	0.5020	0	0.4980	0
0.1250	0	0.7500	0	0.1250
0	0.4980	0	0.5020	0
0.1230	0	0.7500	0	0.1270

P^{∞} =

0.1250	0	0.7500	0	0.1250
0	0.5000	0	0.5000	0
0.1250	0	0.7500	0	0.1250
0	0.5000	0	0.5000	0
0.1250	0	0.7500	0	0.1250

11.14 (a) There are $p+1$ states (0 to p).

The state diagram is:

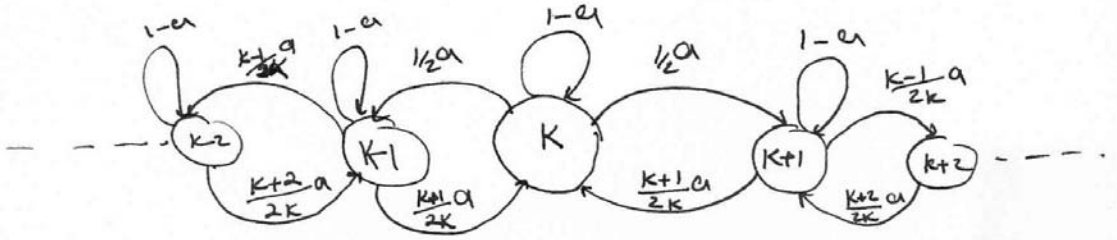


This model is the same as 11.3_d when $a=1$ and $K = p$

* Another way to interpret this problem is to assume that we have one container but to distinguish between particles of container 2 (from 1) we use color for particles, white for container 1 and black for container 2 particles. This converts the problem to 11.3.d with $a=1$ (because when a particle is chosen it will definitely converts its color)

(b) "Central Force"

$p = 2K$



This diagram shows that when the system is not in the central state (K), the probability of moving toward state k is increasing as the system state goes to the left (right).

for example $\begin{cases} p_{k-1,k} = \frac{k+1}{2k}a, \quad p_{k-2,k-1} = \frac{k+2}{2k}a \text{ ---} \\ p_{k,k+1} = \frac{1}{2}a, \quad p_{k+1,k+2} = \frac{k-1}{2k}a, \quad p_{k+2,k+3} = \frac{k-2}{2k}a \text{ ---} \end{cases}$

This shows that the system exhibits a force towards central state.

(c) * to unify notation with Problem 11.3.d we set $p=K$

$x=(x_0, x_1, \cdots, x_K)$ is pmf (stationary)

$$xP = x \Rightarrow \frac{1}{K}x_1 = x_0 \Rightarrow x_1 = Kx_0 = \binom{K}{1}x_0$$

$$x_0 + \frac{2}{K}x_2 = x_1 \Rightarrow x_2 = \frac{K(K-1)}{2}x_0 = \binom{K}{2}x_0$$

$$\frac{K-1}{K}x_1 + \frac{3}{K}x_3 = x_2$$

$$\vdots$$

$$\frac{K-i}{K}x_i + \frac{i+2}{K}x_{i+2} = x_{i+1}$$

$$x_3 = K\left(\frac{K(K-1)}{2} - \frac{K-1}{K}\times K\right)x_0 \Rightarrow x_3 = \frac{K(K-1)(K-2)}{3!}x_0 = \binom{K}{3}x_0$$

we claim $x_i = \binom{K}{i}x_0 = \frac{K!}{i!(K-i)!}x_0$

by induction we prove our claim. for $i=1,2,3$ it is already shown

we assume that it is valid for i & $(i+1)$ then:

$$\frac{K-i}{K}x_i + \frac{i+2}{K}x_{i+2} = x_{i+1}$$

$$\frac{i+2}{K}x_{i+2} = \frac{K!}{(i+1)!(K-i-1)!}x_0 - \frac{K-i}{K}\times\frac{K!}{i!(K-i)!}x_0$$

$$\frac{i+2}{K}x_{i+2} = \frac{(K-1)!}{i!(K-i-1)!}\left(\frac{K}{i+1} - 1\right)x_0 = \frac{(K-1)!}{(i+1)!(K-i-2)!}x_0$$

$$\Rightarrow x_{i+2} = \frac{K!}{(i+2)!(K-i-2)!}x_0 = \binom{K}{i+2}x_0$$

on the other hand $\sum_{i=0}^{K} x_i = 1 \Rightarrow \sum_{i=0}^{K}\binom{K}{i}x_0 = 1$

$$\Rightarrow x_0 = \frac{1}{\sum_{i=0}^{K}\binom{K}{i}} = \frac{1}{2^K} \quad (K=P)$$

$$\Rightarrow x = \frac{1}{2^P}\left(\binom{P}{0}, \binom{P}{1}, \binom{P}{2}, \cdots, \binom{P}{P-1}, \binom{P}{P}\right)$$

11.15 (a)

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{1}{16} & \frac{6}{16} & \frac{9}{16} & 0 & 0 \\ 0 & \frac{4}{16} & \frac{8}{16} & \frac{4}{16} & 0 \\ 0 & 0 & \frac{4}{16} & \frac{6}{16} & \frac{1}{16} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

11.15 b, c

P =

0	1.0000	0	0	0
0.0625	0.3750	0.5625	0	0
0	0.2500	0.5000	0.2500	0
0	0	0.5625	0.3750	0.0625
0	0	0	1.0000	0

P^2 =

0.0625	0.3750	0.5625	0	0
0.0234	0.3438	0.4922	0.1406	0
0.0156	0.2188	0.5313	0.2188	0.0156
0	0.1406	0.4922	0.3438	0.0234
0	0	0.5625	0.3750	0.0625

P^4 =

0.0215	0.2754	0.5186	0.1758	0.0088
0.0172	0.2544	0.5131	0.2043	0.0110
0.0144	0.2280	0.5151	0.2280	0.0144
0.0110	0.2043	0.5131	0.2544	0.0172
0.0088	0.1758	0.5186	0.2754	0.0215

P^8 =

0.0147	0.2317	0.5143	0.2254	0.0139
0.0145	0.2301	0.5143	0.2270	0.0141
0.0143	0.2286	0.5143	0.2286	0.0143
0.0141	0.2270	0.5143	0.2301	0.0145
0.0139	0.2254	0.5143	0.2317	0.0147

P^{∞} =

0.0143	0.2286	0.5143	0.2286	0.0143
0.0143	0.2286	0.5143	0.2286	0.0143
0.0143	0.2286	0.5143	0.2286	0.0143
0.0143	0.2286	0.5143	0.2286	0.0143
0.0143	0.2286	0.5143	0.2286	0.0143

11.16 (a) It is easy to see that this problem is equivalent to 11.4.c in fact

$$P(X_{n+1}=m \mid X_n=m) = \frac{m}{p}\times\frac{p-m}{p} + \frac{p-m}{p}\times\frac{m}{p} = \frac{2m(p-m)}{p^2}$$

$$P(X_{n+1}=m-1 \mid X_n=m) = \frac{m}{p}\times\frac{m}{p} = \frac{m^2}{p^2}$$

$$P(X_{n+1}=m+1 \mid X_n=m) = 1 - \frac{m^2}{p^2} - \frac{2m(p-m)}{p^2} = \frac{(p-m)^2}{p^2}$$

The same as in 11.4.c

(b)

$$P = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ \frac{1}{p^2} & \frac{2(p-1)}{p^2} & \frac{(p-1)^2}{p^2} & \cdots & 0 \\ 0 & \frac{4}{p^2} & \frac{4(p-2)}{p^2} & \frac{(p-2)^2}{p^2} & \cdots \\ \vdots & & \ddots & & \\ & & & \frac{(p-1)^2}{p^2} & \frac{2(p-1)}{p^2} \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix} \qquad x=(x_0, x_1, \cdots, x_p)$$

$$xP=x \Rightarrow \frac{x_1}{p^2} = x_0 \Rightarrow x_1 = p^2 x_0 = \binom{p}{1}^2 x_0$$

$$x_0 + \frac{2(p-1)}{p^2}x_1 + \frac{4}{p^2}x_2 = x_1 \Rightarrow x_2 = \frac{1}{4}(p-1)^2 x_1 = \frac{1}{4}p^2(p-1)^2 x_0 = \binom{p}{2}^2 x_0$$

$$\frac{(p-1)^2}{p^2}x_1 + 4\frac{(p-2)}{p^2}x_2 + \frac{9}{p^2}x_3 = x_2 \Rightarrow x_3 = \frac{1}{9}(p-2)^2 x_2 = \frac{1}{9}\times\frac{1}{9}p^2(p-1)^2(p-2)^2 x_0$$

$$\Rightarrow x_3 = \binom{p}{3}^2 x_0$$

Although we can obtain the general form $x_m = \binom{p}{m}^2 x_0$ by induction, we take another approach. As the Markov chain here is reversible: $x_i P_{ij} = x_j P_{ji} \Rightarrow x_i P_{i,i+1} = x_{i+1} P_{i+1,i}$

$$\Rightarrow x_i \frac{(p-i)^2}{p^2} = x_{i+1}\frac{(i+1)^2}{p^2} \Rightarrow x_{i+1} = \frac{(p-i)^2}{(i+1)^2}x_i$$

$$x_{i+1} = \frac{(p-i)^2(p-i+1)^2(p-(i+2))^2 \cdots p^2}{(i+1)^2 i^2 (i-1)^2 \cdots 1}x_0 = \left(\frac{(p-i)\times(p-i+1)\times\cdots\times p}{(i+1)!}\right)^2 x_0 = \left(\frac{p!}{(i+1)!(p-i-1)!}\right)^2 x_0 = \binom{p}{i+1}^2 x_0$$

11.16 (b) (Cont.)

$$x_{i+1} = \binom{p}{i+1}^2 x_0 \qquad \text{we know} \quad \sum_{i=0}^{p} x_i = 1$$

$$\Rightarrow x_0 = \frac{1}{\sum_{i=0}^{p} \binom{p}{i}^2} = \frac{1}{\binom{2p}{p}}$$

$$\Rightarrow x_i = \frac{\binom{p}{i}^2}{\binom{2p}{p}}$$

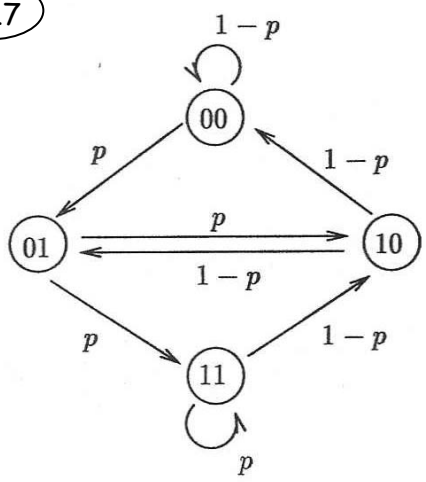
We have used the fact that $\sum_{i=0}^{p} \binom{p}{i}^2 = \binom{2p}{p}$

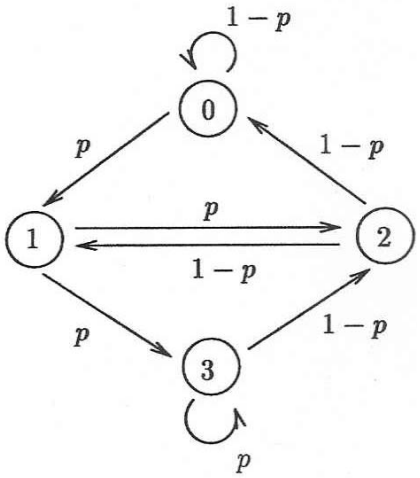
to see this, we can use $\binom{p}{i} = \binom{p}{p-i}$

$$\Rightarrow \sum_{i=0}^{p} \binom{p}{i}^2 = \sum_{i=0}^{p} \binom{p}{i}\binom{p}{p-i} = \binom{2p}{p}$$

which follows if we note that both sides show the number of subgroups of size p one can select from a set of n white balls and n black balls.

11.17





a) $\mathbf{P} = \begin{bmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \\ 0 & 0 & q & p \end{bmatrix}$

b) $\mathbf{P}^2 = \begin{bmatrix} q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \end{bmatrix}$

(01) to (01) in 2 steps corresponds to $p_{11}(2) = pq$, i.e.

$$p(2|1)p(1|2) = qp \quad \surd$$

c) Let A be a matrix with rows $\underline{r}_1, \underline{r}_2, \underline{r}_3, \underline{r}_4$, then

$$\mathbf{P}A = \begin{bmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \\ 0 & 0 & q & p \end{bmatrix} \begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \\ \underline{r}_3 \\ \underline{r}_4 \end{bmatrix} = \begin{bmatrix} q\underline{r}_1 + p\underline{r}_2 \\ q\underline{r}_3 + p\underline{r}_4 \\ q\underline{r}_1 + p\underline{r}_2 \\ q\underline{r}_3 + p\underline{r}_4 \end{bmatrix}$$

But all rows of $\mathbf{P}^2$ are equal, thus

$$\mathbf{P}\mathbf{P}^2 = \begin{bmatrix} q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \\ q^2 & qp & qp & p^2 \end{bmatrix} = \mathbf{P}^2$$

and if we assume $\mathbf{P}^{n-1} = \mathbf{P}^2$. Then $\mathbf{P}^n = \mathbf{P}\mathbf{P}^{n-1} = \mathbf{P}\mathbf{P}^2 = \mathbf{P}^2$

$$\therefore \mathbf{P}^n = \mathbf{P}^2 \quad n \geq 2$$

After two steps, the process is independent of the post.

d) Let $\underline{p}(0)$ be the initial state post, then

$$\underline{p}(0)\mathbf{P}^n = (q^2, qp, qp, p^2) \text{ for } n \geq 2$$

11.18

(a) IF $X_n = i$ then X_{n+1} could be either $i+1$, or 0

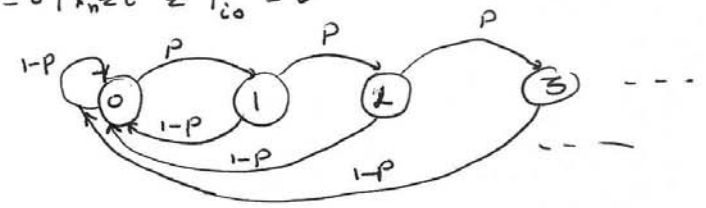
SFSSFFSSSS|S| $\rightarrow X_{n+1}=5$ ($X_n=4$)

SFSSFFSSSS|F| $\rightarrow X_{n+1}=0$ ($X_n=4$)

This is independent of the events/states before X_{n+1} $\Rightarrow$ Markov

(b) $P(X_{n+1}=i+1 / X_n=i) = P_{i,i+1} = p$ → probability of success

$P(X_{n+1}=0 \mid X_n=i) = P_{i0} = 0$



(c)
$$P = \begin{bmatrix} 1-p & p & 0 & 0 & 0 & \cdots \\ 1-p & 0 & p & 0 & 0 & \cdots \\ 1-p & 0 & 0 & p & & \\ 1-p & 0 & 0 & 0 & p & \ddots \\ \vdots & \vdots & \vdots & & & \end{bmatrix} \qquad \pi P = \pi$$

$\Rightarrow \quad \pi_0 = (1-p)(\pi_0 + \pi_1 + \pi_2 + \cdots)$

$\pi_1 = p\pi_0$

$\pi_2 = p\pi_1 = p^2\pi_0$

$\pi_3 = p\pi_2 = p^3\pi_0$

$\vdots$

$\pi_n = p^n\pi_0$

$$\sum_{i=0}^{\infty} \pi_i = 1 \Rightarrow (1 + p + p^2 + \cdots)\pi_0 = 1$$

$p<1 \Rightarrow \frac{1}{1-p}\pi_0 = 1 \Rightarrow \pi_0 = 1-p$

$$\pi = \left(1-p \quad p(1-p) \quad p^2(1-p) \quad \cdots \right)$$

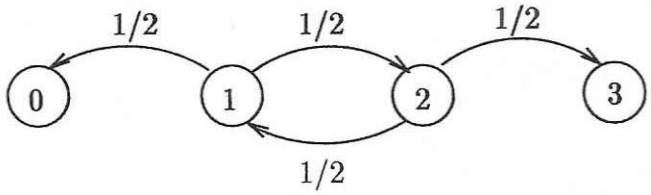
$$\pi_n = p^n(1-p)$$

11.19

S.11 $X_n \epsilon 0, 1, 2, 3$

a) $P[X_n = k | X_{n-1} = j, ...] = P[X_n = k | X_{n-1} = j]$
since $X_n = X_{n-1} \pm 1$ for $X_{n-1} \epsilon \{1, 2\}$
and $X_n = X_{n-1}$ if $X_{n-1} \epsilon \{0, 3\}$

b)



$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c) for $n = 2k$, $i \epsilon \{1, 2\}$

$$\begin{aligned} p_{ii}(n) &= P[\overbrace{HT \;\; HT \;\; HT \;\; ... \;\; HT}^{2k}] = \left(\frac{1}{2}\right)^n \\ p_{10}(n) &= \sum_{j=0}^{k-1} P[j \; 1 \to 2 \text{ cycles and then go to } 0] \\ &= \sum_{j=0}^{k-1} \left(\frac{1}{2}\right)^{2j} \frac{1}{2} \\ &= \frac{2}{3}\left(1 - \left(\frac{1}{4}\right)^k\right) \\ p_{23}(n) &= p_{10}(n) \quad \text{by symmetry.} \end{aligned}$$

d) $$\mathbf{P}(n) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{2}{3}(1 - (\frac{1}{4})^k) & (\frac{1}{2})^n & 0 & \frac{1}{3}(1 - (\frac{1}{4})^k \\ \frac{1}{3}(1 - (\frac{1}{4})^k) & 0 & (\frac{1}{2})^n & \frac{2}{3}(1 - (\frac{1}{4})^k) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

e) $\mathbf{P}(n) \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$

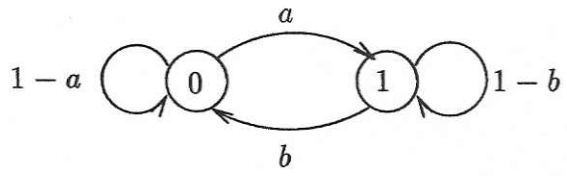
f) $$\begin{aligned} \underline{p}(n) &= [0100]\mathbb{P}(n) \\ &= \left[\frac{2}{3}\left(1-\left(\frac{1}{4}\right)^k\right), \left(\frac{1}{4}\right)^k, 0, \frac{1}{3}\left(1-\left(\frac{1}{4}\right)^k\right)\right] \\ &\rightarrow \left[\frac{2}{3}, 0, 0, \frac{1}{3}\right] \end{aligned}$$

$P[\text{player } A \text{ wins}] = \frac{1}{3}$.

11.20

8.12 $X_n \epsilon \{0,1\}$ where 0 = working, 1 = not working

a) $\mathbf{P} = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$



b) To find the eigenvalues, consider

$$\begin{aligned} |\mathbf{P} - \lambda I| &= (1-b-\lambda)(1-a-\lambda) - ab = 0 \\ &\Rightarrow \quad \lambda_1 = 1 \qquad \lambda_2 = 1-a-b \end{aligned}$$

Then the eigenvectors are $\underline{e}_1 = [1, \frac{b}{a}]$, $\underline{e}_2 = [1, -1]$, so

$$E = \begin{bmatrix} \underline{e}_1 \\ \underline{e}_2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{b}{a} \\ 1 & -1 \end{bmatrix}$$

and

$$E^{-1} = \frac{1}{a+b} \begin{bmatrix} a & b \\ a & -a \end{bmatrix}$$

and thus

$$\begin{aligned} \mathbf{P}^n &= E^{-1} \begin{bmatrix} 1 & 0 \\ 0 & (1-a-b)^n \end{bmatrix} E \\ &= \frac{1}{a+b} \begin{bmatrix} a+b(1-a-b)^n & b-b(1-a-b)^n \\ a-a(1-a-b)^n & b+c(1-a-b)^n \end{bmatrix} \end{aligned}$$

c) $0 < a+b < 2$ since $0 < a < 1$ and $0 < b < 1$
$\Rightarrow -1 < 1-a-b < 1 \Rightarrow (1-a-b)^n \to 0$

$$\therefore \mathbf{P}^n \to \begin{bmatrix} \frac{a}{a+b} & \frac{b}{a+b} \\ \frac{a}{a+b} & \frac{b}{a+b} \end{bmatrix}$$

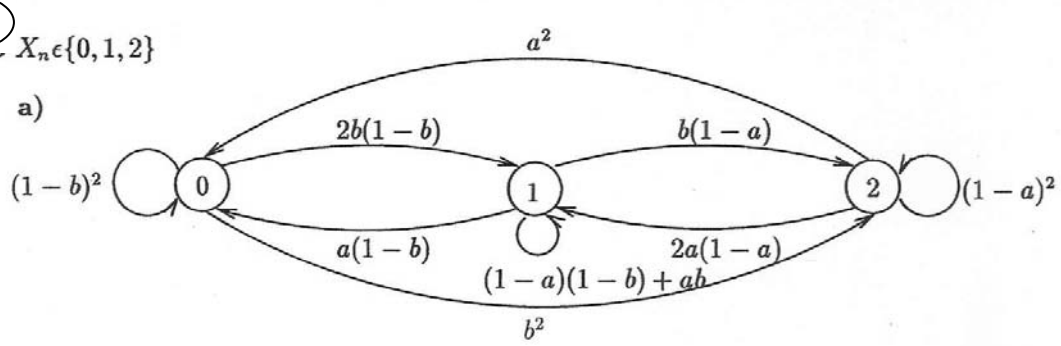
and

$$\underline{p}(n) \to [\frac{a}{a+b}, \frac{b}{a+b}]$$

11.21

$X_n \epsilon \{0, 1, 2\}$

a)



$$\mathbf{P} = \begin{bmatrix} (1-b)^2 & 2b(1-b) & b^2 \\ a(1-b) & (1-a)(1-b)+ab & b(1-a) \\ a^2 & 2a(1-a) & (1-a)^2 \end{bmatrix}$$

b) Claim: the steady state pmf is

$$\underline{p} = \left(\left(\frac{a}{a+b}\right)^2, 2\left(\frac{b}{a+b}\right)\left(\frac{a}{a+b}\right), \left(\frac{b}{a+b}\right)^2 \right)$$
$$= \frac{1}{(a+b)^2}(a^2, 2ab, b^2)$$

$$\underline{p}\mathbf{P} = \frac{1}{(a+b)^2}(a^2, 2ab, b^2) \begin{bmatrix} (1-b)^2 & 2b(1-b) & b^2 \\ a(1-b) & (1-a)(1-b)+ab & b(1-a) \\ a^2 & 2a(1-a) & (1-a)^2 \end{bmatrix}$$
$$= \frac{1}{(a+b)^2} \begin{bmatrix} a^2(1-b)^2 + 2a^2b(1-b) + a^2b^2 \\ 2a^2b(1-b) + 2ab(1-a)(1-b) + 2a^2b^2 + 2ab^2(1-a) \\ a^2b^2 + 2ab^2(1-a) + b^2(1-a)^2 \end{bmatrix}^+$$
$$= \frac{1}{(a+b)^2}(a^2, 2ab, b^2) = \underline{p} \quad \surd$$

c) In the general case with n machines, the steady state part is given by

$$P[X_n = k] = \binom{n}{k} \left(\frac{a}{a+b}\right)^k \left(\frac{b}{a+b}\right)^{n-k} \qquad 0 \le k \le n$$

11.22

(a) It is a direct result of eq. (11.9)

(11.9) $\Rightarrow$ $1 = \sum_j P(X_{n+1}=j \mid X_n=i) = \sum_j P_{ij}$ $\Rightarrow$ Transition Matrix is stochastic

(b) Consider P_i as the ith row of P and φ_j as the jth column of Φ:

$$P_i = (P_{i1}, P_{i2}, \cdots, P_{in}), \quad \varphi_j = (\varphi_{1j}, \varphi_{2j}, \cdots \varphi_{nj})^t, \quad \underline{1} = (1,1,1,\cdots,1)^t$$

$$P\Phi = \begin{pmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{pmatrix} (\varphi_1 \varphi_2 \cdots \varphi_n) = \begin{pmatrix} P_1\varphi_1 & P_1\varphi_2 & \cdots & P_{1n} \\ P_2\varphi_1 & P_2\varphi_2 & \cdots & P_{2n} \\ \vdots & & & \\ P_n\varphi_1 & P_n\varphi_2 & \cdots & P_n\varphi_n \end{pmatrix}$$

for kth row $\sum_{j=1}^{n} P_k \varphi_j = P_k \sum_{j=1}^{n} \varphi_j = P_k \underline{1} = \sum_j P_{kj} = 1 \Rightarrow P\Phi$ stochastic

(C) P^2 is clearly stochastic (direct result of b with $P=\Phi$)

by induction if $\Phi = P^n$ is stochastic then:

$$P^{n+1} = P \cdot P^n = P \cdot \Phi \xRightarrow{\text{based on b}} P^{n+1} \text{ is stochastic}$$

11.23

~~8.10~~ If $\mathbf{P}^k = \begin{bmatrix} \underline{r} \\ \underline{r} \\ \vdots \\ \underline{r} \end{bmatrix}$, that is, $\mathbf{P}^k$ has identical rows

then the jth row of $\mathbf{P}^{k+1}$ is

$$\begin{aligned} \mathbf{P}\mathbf{P}^k &= \begin{bmatrix} & & \cdots & \\ p_{j1} & p_{j2} & \cdots & p_{jn} \\ & & \cdots & \end{bmatrix} \begin{bmatrix} \underline{r} \\ \underline{r} \\ \vdots \\ \underline{r} \end{bmatrix} \\ &= \begin{bmatrix} & & \cdots & \\ p_{j1}\underline{r} + p_{j2}\underline{r} + \ldots + p_{jn}\underline{r} \\ & & \cdots & \end{bmatrix} \\ &= \begin{bmatrix} \cdots \\ \underline{r} \\ \cdots \end{bmatrix} = \mathbf{P}^k \end{aligned}$$

11.24

$P(n+m) = P(n)P(m)$

$\Rightarrow P(2) = P(1) \times P(1) = P^2$

if $P(n) = P^n$ by induction

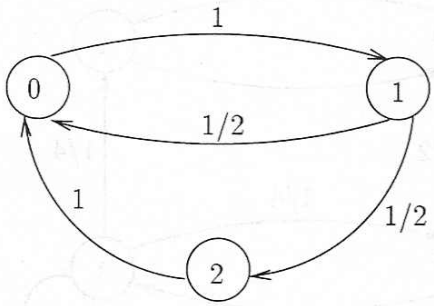
then $P(n+1) = P(n) \times P(1) = P^n \times P = P^{n+1}$

11.3 Classes of States, Recurrence Properties, and Limiting Probabilities

11.25

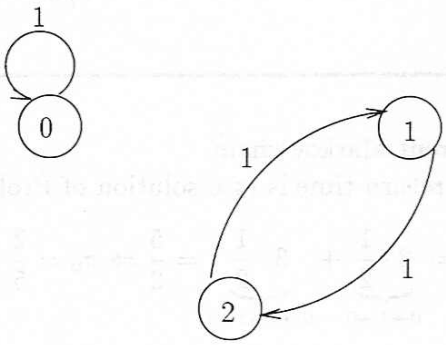
(a)

(i)



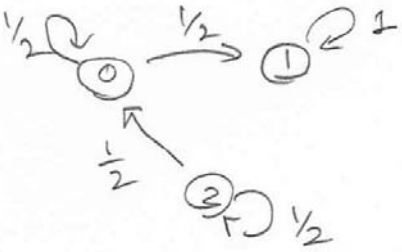
one class {1,2,3}
recurrent

(ii)



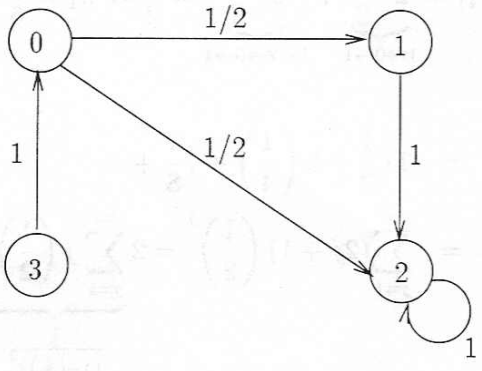
{0} recurrent
{2,3} recurrent

(iii)



{2}, {0} transient
{1} recurrent

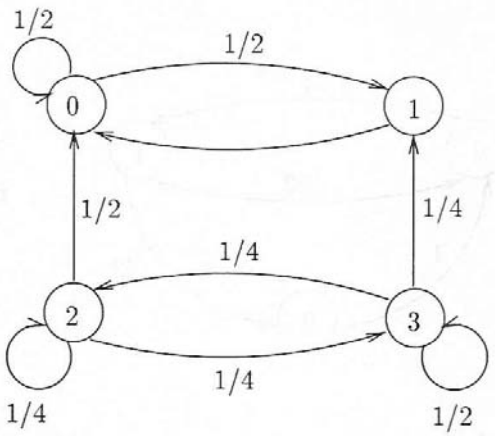
(iv)



{0} transient
{1} transient
{2} recurrent
{3} transient

(c)

(iv)



{0,1} recurrent

{2,3} transient

(i)

$$A^2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad A^4 = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/2 & 1/4 & 1/4 \\ 1/2 & 1/2 & 0 \end{bmatrix} \quad A^{50} = \begin{bmatrix} .4 & .4 & .2 \\ .4 & .4 & .2 \\ .4 & .4 & .2 \end{bmatrix}$$

(ii)

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A^3 = A \quad A^4 = I \quad A^5 = A \quad \ldots$$

alternates

(iii)

$$A^2 = \begin{bmatrix} .25 & .75 & 0 \\ 0 & 1 & 0 \\ .5 & .25 & .25 \end{bmatrix} \quad A^4 = \begin{bmatrix} .0625 & .9375 & 0 \\ 0 & 1 & 0 \\ .25 & 0.6875 & 0.0625 \end{bmatrix} \quad A^{50} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(iv)

$$A^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1/2 & 1/2 & 0 \end{bmatrix} \quad A^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

converges in 3 steps.

(v)

$$A^2 = \begin{bmatrix} 3/4 & 1/4 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 3/8 & 5/16 & 1/8 & 3/16 \\ 3/8 & 1/8 & 3/16 & 5/16 \end{bmatrix} \quad A^8 = \begin{bmatrix} .668 & .332 & 00 \\ .664 & .336 & 0 & 0 \\ .648 & .327 & .009 & .015 \\ .639 & .321 & .015 & .0244 \end{bmatrix} \quad A^{50} = \begin{bmatrix} 2/3 & 1/3 & 0 & 0 \\ 2/3 & 1/3 & 0 & 0 \\ 2/3 & 1/3 & 0 & 0 \\ 2/3 & 1/3 & 0 & 0 \end{bmatrix}$$

11.26

8.21 a) This is a positive recurrent Markov chain.
For state 0 the mean first return time is (see solution of Problem 11.25)

$$\mathcal{E}[T_0] = \underbrace{2 \cdot \frac{1}{2}}_{0\to1\to0} + \underbrace{3 \cdot \frac{1}{2}}_{0\to1\to2\to0} = \frac{5}{2} \Rightarrow \pi_0 = \frac{2}{5}$$

For state 1

$$\mathcal{E}[T_1] = \underbrace{2 \cdot \frac{1}{2}}_{1\to0\to1} + \underbrace{3 \cdot \frac{1}{2}}_{1\to2\to0\to1} = \frac{5}{2} \Rightarrow \pi_1 = \frac{2}{5}$$

For state 2

$$\begin{aligned}\mathcal{E}[T_2] &= 3 \cdot \frac{1}{2} + 5\left(\frac{1}{4}\right) + 7\frac{1}{8} + \ldots \\ &= \sum_{j=1}^{\infty}(2j+1)\left(\frac{1}{2}\right)^j = 2\underbrace{\sum_{j=1}^{\infty} j\left(\frac{1}{2}\right)^j}_{\frac{\frac{1}{2}}{\left(1-\left(\frac{1}{2}\right)\right)^2}}\end{aligned}$$

$$\Rightarrow \pi_2 = \frac{1}{5}$$

It can easily be shown that above are solutions to equations for the stationary proof.

b) If system starts in state 0 then $\mathcal{E}[T_0] = 1 \Rightarrow \pi_0 = 1$
If system starts in state 1 or 2 then $\mathcal{E}[T_1] = \mathcal{E}[T_2] = 2$

$$\Rightarrow \pi_1 = \pi_2 = \tfrac{1}{2}$$

c) Only state 2 is recurrent and $\mathcal{E}[T_2] = 1 \Rightarrow \pi_2 = 1$

d) States 0 and 1 are recurrent and

$$\begin{aligned}\mathcal{E}[T_0] &= 1 \cdots \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2} \\ \Rightarrow \pi_0 &= \frac{2}{3} \\ \Rightarrow \pi_1 &= \frac{1}{3}\end{aligned}$$

(11.26)

(i) This is an irreducible positive recurrent mc.

$$(p_0, p_1, p_2) = (p_0, p_1, p_2)\begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \end{bmatrix}$$

$$p_0 = \tfrac{1}{2}p_1 + p_2$$

$$p_1 = p_0$$

$$p_2 = \tfrac{1}{2}p_1$$

$$1 = p_0 + p_1 + p_2 = p_0 + p_0 + \tfrac{1}{2}p_0 = \tfrac{5}{2}p_0$$

$$p_0 = \tfrac{2}{5} = p_1$$

$$p_2 = \tfrac{1}{5}$$

(ii) This is a multiclass mc

Class 1 consists of a single recurrent state

Class 2 is has 2 recurrent states and has period 2

Suppose we calculate the stationary pmf:

$p_0 = p_0$ $\quad p_0 = p_0$ $\quad p_0 + 2p_1 = 1$

$p_1 = p_2$ $\quad p_1 = p_2$ $\quad$ No unique solution

$p_2 = p_1$

Let $p_0 = \alpha$

then $p_1 = p_2 = \frac{1-\alpha}{2}$

$$(p_0, p_1, p_2) = \left(\alpha, \tfrac{1-\alpha}{2}, \tfrac{1-\alpha}{2}\right)$$

$$= \alpha(1,0,0) + (1-\alpha)\left(0, \tfrac{1}{2}, \tfrac{1}{2}\right)$$

$\alpha(1,0,0)$: stationary pmf for class 1 in isolation

$(1-\alpha)\left(0, \frac{1}{2}, \frac{1}{2}\right)$: stationary pmf for class 2 in isolation

(iii)

$$\pi_0 = \tfrac{1}{2}\pi_0 + \pi_2$$
$$\pi_1 = \tfrac{1}{2}\pi_0 + \pi_1 \Rightarrow \pi_0 = 0$$
$$\pi_2 = \tfrac{1}{2}\pi_0 + \tfrac{1}{2}\pi_2 \Rightarrow \pi_2 = 0$$

$\pi_0 + \pi_1 + \pi_2 = 1$
$\Rightarrow \pi_1 = 1$

This is a multiclass MC with 2 transient classes and one recurrent single state class, $\{1\}$.

(iv)

$$\pi_0 = \pi_3$$
$$\pi_1 = \tfrac{1}{2}\pi_0$$
$$\pi_2 = \tfrac{1}{2}\pi_0 + \pi_1 + \pi_2$$
$$\pi_3 = 0 \Rightarrow \pi_3 = 0 \Rightarrow \pi_1 = 0 \Rightarrow \pi_2 = 1$$

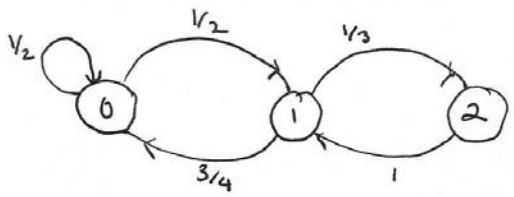
This MC has 3 transient classes and one recurrent, single state class.

(v)

$$\pi_0 = \tfrac{1}{2}\pi_0 + \pi_1 + \tfrac{1}{2}\pi_2 \qquad \tfrac{1}{2}\pi_0 = \pi_1 \Rightarrow \pi_0 = \tfrac{2}{3}, \quad \pi_1 = \tfrac{1}{3}$$
$$\pi_1 = \tfrac{1}{2}\pi_0 + \tfrac{1}{4}\pi_3$$
$$\pi_2 = \tfrac{1}{4}\pi_2 + \tfrac{1}{4}\pi_3 \Rightarrow \pi_3 = 3\pi_2$$
$$\pi_3 = \tfrac{1}{4}\pi_2 + \tfrac{1}{2}\pi_3 \Rightarrow \pi_3 = \tfrac{1}{2}\pi_2 \Rightarrow \pi_2 = \pi_3 = 0$$

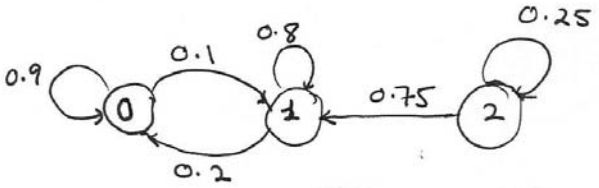
This MC has a transient class and a two-state, aperiodic, positive recurrent class.

11.27 (a)



irreducible

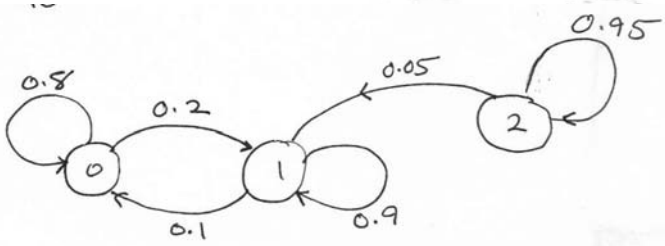
(b)



{2} transient

{0,1} recurrent

(c)



11.28 (a)

i)

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

ii)

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

iii)

$$P_2 = \begin{array}{c} \begin{matrix} 0 & 1 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 \end{pmatrix} \end{array}$$

iv

$$P_2 = \begin{array}{c} \begin{matrix} 0 & 1 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 \end{pmatrix} \end{array}$$

(b) (i) Recurrent, Periodic

(ii) Recurrent, Periodic

(iii) Recurrent. Periodic

(iv) Transient, aperiodic

(c) left eigenvector related to eigenvalue 1 for these Markov chains are

(i) $\pi = (0.1667 \quad 0.1667 \quad 0.3333 \quad 0.1667 \quad 0.1667)^t$

(ii) $\pi = (0.1429 \quad 0.1429 \quad 0.2857 \quad 0.1429 \quad 0.1429)^t$

(iii) $\pi = (0.1667 \quad 0.1667 \quad 0.3333 \quad 0.3333)^t$

(iv) $\pi = (0 \quad 1 \quad 0 \quad 0)^t$

(d)

P_i^{∞} =

0	0.5000	0	0	0.5000
0	0	1.0000	0	0
0.5000	0	0	0.5000	0
0	0.5000	0	0	0.5000
0	0	1.0000	0	0

P_{ii}^{∞} =

0.1429	0.1429	0.2857	0.1429	0.1429	0.1429
0.1429	0.1429	0.2857	0.1429	0.1429	0.1429
0.1429	0.1429	0.2857	0.1429	0.1429	0.1429
0.1429	0.1429	0.2857	0.1429	0.1429	0.1429
0.1429	0.1429	0.2857	0.1429	0.1429	0.1429
0.1429	0.1429	0.2857	0.1429	0.1429	0.1429

P_{iii}^{∞} =

0.3333	0	0.6667	0
0	0.3333	0	0.6667
0.3333	0	0.6667	0
0	0.3333	0	0.6667

P_{iv}^{∞} =

0.0000	0.6000	0.0000	0
0	1.0000	0	0
0.0000	0.6000	0.0000	0
0	0.4000	0	0.0000

11.29

```
alpha=0.85;
P1=[0 1 0 0 0
    0 0 1 0 0
    0.5 0 0 0.5 0
    0 0 0 0 1
    0 0 1 0 0];
k=length(P1);
R1=alpha*P1+(1-alpha)*1/k*ones(k,k);
R11=R1-eye(k,k);
Q1=[ones(k,1) R11(:,2:k)];
Ps1=[1 zeros(1,k-1)]*inv(Q1);

P2=[0 1 0 0 0 0
    0 0 1 0 0 0
    0.5 0 0 0.5 0 0
    0 0 0 0 1 0
    0 0 0 0 0 1
    0 0 1 0 0 0];
k=length(P2);
R2=alpha*P2+(1-alpha)*1/k*ones(k,k);
R22=R2-eye(k,k);
Q2=[ones(k,1) R22(:,2:k)];
Ps2=[1 zeros(1,k-1)]*inv(Q2);

P3=[0 1/3 0 2/3
    1/3 0 2/3 0
    0 1/3 0 2/3
    1/3 0 2/3 0];
k=length(P3);
R3=alpha*P3+(1-alpha)/k*ones(k,k);
R33=R3-eye(k,k);
Q3=[ones(k,1) R33(:,2:k)];
Ps3=[1 zeros(1,k-1)]*inv(Q3);

P4=[0 1/3 0 2/3
   0 1 0 0
   0 1/3 0 2/3
   1/3 0 2/3 0];
k=length(P4);
R4=alpha*P4+(1-alpha)*1/k*ones(k,k);
R44=R4-eye(k,k);
Q4=[ones(k,1) R44(:,2:k)];
Ps4=[1 zeros(1,k-1)]*inv(Q4);
```

```
P1 =

         0    1.0000         0         0         0
         0         0    1.0000         0         0
    0.5000         0         0    0.5000         0
         0         0         0         0    1.0000
         0         0    1.0000         0         0

>> R1

R1 =

    0.0300    0.8800    0.0300    0.0300    0.0300
    0.0300    0.0300    0.8800    0.0300    0.0300
    0.4550    0.0300    0.0300    0.4550    0.0300
    0.0300    0.0300    0.0300    0.0300    0.8800
    0.0300    0.0300    0.8800    0.0300    0.0300

>> Q1

Q1 =

    1.0000    0.8800    0.0300    0.0300    0.0300
    1.0000   -0.9700    0.8800    0.0300    0.0300
    1.0000    0.0300   -0.9700    0.4550    0.0300
    1.0000    0.0300    0.0300   -0.9700    0.8800
    1.0000    0.0300    0.8800    0.0300   -0.9700

>> Ps1

Ps1 =

    0.1670    0.1719    0.3223    0.1670    0.1719

>> P2

P2 =

         0    1.0000         0         0         0         0
         0         0    1.0000         0         0         0
    0.5000         0         0    0.5000         0         0
         0         0         0         0    1.0000         0
         0         0         0         0         0    1.0000
         0         0    1.0000         0         0         0

>> R2

R2 =

    0.0250    0.8750    0.0250    0.0250    0.0250    0.0250
    0.0250    0.0250    0.8750    0.0250    0.0250    0.0250
    0.4500    0.0250    0.0250    0.4500    0.0250    0.0250
```

```
    0.0250    0.0250    0.0250    0.0250    0.8750    0.0250
    0.0250    0.0250    0.0250    0.0250    0.0250    0.8750
    0.0250    0.0250    0.8750    0.0250    0.0250    0.0250

>> Q2

Q2 =

    1.0000    0.8750    0.0250    0.0250    0.0250    0.0250
    1.0000   -0.9750    0.8750    0.0250    0.0250    0.0250
    1.0000    0.0250   -0.9750    0.4500    0.0250    0.0250
    1.0000    0.0250    0.0250   -0.9750    0.8750    0.0250
    1.0000    0.0250    0.0250    0.0250   -0.9750    0.8750
    1.0000    0.0250    0.8750    0.0250    0.0250   -0.9750

>> Ps2

Ps2 =

    0.1421    0.1458    0.2755    0.1421    0.1458    0.1489

>> P3

P3 =

         0    0.3333         0    0.6667
    0.3333         0    0.6667         0
         0    0.3333         0    0.6667
    0.3333         0    0.6667         0

>> R3

R3 =

    0.0375    0.3208    0.0375    0.6042
    0.3208    0.0375    0.6042    0.0375
    0.0375    0.3208    0.0375    0.6042
    0.3208    0.0375    0.6042    0.0375

>> Q3

Q3 =

    1.0000    0.3208    0.0375    0.6042
    1.0000   -0.9625    0.6042    0.0375
    1.0000    0.3208   -0.9625    0.6042
    1.0000    0.0375    0.6042   -0.9625

>> Ps3

Ps3 =
```

```
    0.1792    0.1792    0.3208    0.3208

>> P4

P4 =

         0    0.3333         0    0.6667
         0    1.0000         0         0
         0    0.3333         0    0.6667
    0.3333         0    0.6667         0

>> R4

R4 =

    0.0375    0.3208    0.0375    0.6042
    0.0375    0.8875    0.0375    0.0375
    0.0375    0.3208    0.0375    0.6042
    0.3208    0.0375    0.6042    0.0375

>> Q4

Q4 =

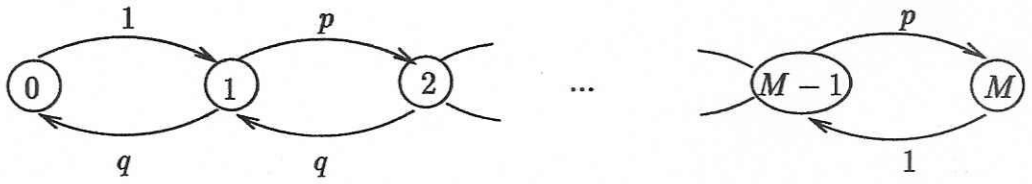
    1.0000    0.3208    0.0375    0.6042
    1.0000   -0.1125    0.0375    0.0375
    1.0000    0.3208   -0.9625    0.6042
    1.0000    0.0375    0.6042   -0.9625

>> Ps4

Ps4 =

    0.0812    0.6395    0.1250    0.1543
```

11.30



This Markov chain has period 2 $\Rightarrow p_{ii}(2n) = 2\pi_i$ as $n \to \infty$

$$\pi_0 = q\pi_1 \quad \Rightarrow \quad \pi_1 = \frac{1}{q}\pi_0$$

$$\pi_1 = \pi_0 + q\pi_2 \quad \Rightarrow \quad \pi_2 = \frac{1}{q}\left(\frac{1}{q} - 1\right)\pi_0 = \frac{p}{q^2}\pi_0$$

$$\pi_2 = p\pi_1 + q\pi_3 \quad \Rightarrow \quad \pi_3 = \frac{1}{q}(\pi_2 - p\pi_1) = \frac{1}{q}\left(\frac{p}{q^2} - \frac{p}{q}\right)\pi_0$$

$$\vdots \qquad = \frac{p}{q^2}\left(\frac{1}{q} - 1\right) = \frac{p^2}{q^3}\pi_0$$

$$\pi_{M-1} = p\pi_{M-2} + q\pi_M \qquad \pi_4 = \frac{1}{q}(\pi_3 - p\pi_2) = \frac{1}{q}\left(\frac{p^2}{q^3} - \frac{p^2}{q^2}\right)\pi_0$$

$$\pi_M = p\pi_{M-1} \qquad = \frac{1}{q}\left(\frac{p^2}{q^2}\right)\left(\frac{1}{q} - 1\right)\pi_0 = \frac{p^3}{q^4}\pi_0$$

$$\vdots$$

$$\text{and} \quad \pi_M = \pi_{M-1} - p\pi_{M-2}$$

$$= \left(\frac{p^{M-2}}{q^{M-1}} - p\frac{p^{M-3}}{q^{M-2}}\right)\pi_0$$

$$= \frac{p^{M-2}}{q^{M-2}}\left(\frac{1}{q}-1\right)\pi_0$$

$$\frac{p^{M-1}}{q^{M-1}}\pi_0$$

To find π_0 we note that

$$\begin{aligned} 1 &= \pi_0\left(1+\frac{1}{q}+\frac{p}{q^2}+\frac{p^2}{q^3}+\ldots+\frac{p^{M-2}}{q^{M-1}}+\frac{p^{M-1}}{q^{M-1}}\right) \\ &= \pi_0\left(1+\frac{1}{q}\left(\underbrace{\left(\frac{p}{q}\right)^0+\left(\frac{p}{q}\right)^1+\ldots+\left(\frac{p}{q}\right)^{M-2}}_{\frac{1-\left(\frac{p}{q}\right)^{M-1}}{1-\frac{p}{q}}}\right)+\frac{p^{M-1}}{q^{M-1}}\right) \end{aligned}$$

$$\therefore \quad \pi_0 = \frac{1-2p}{2\left(1-p\left(1+\left(\frac{p}{q}\right)^{M-1}\right)\right)}$$

and

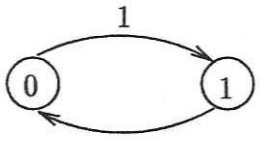
$$\begin{aligned} \pi_i &= \frac{1}{q}\left(\frac{p}{q}\right)^{i-1}\pi_0 \qquad 1 \le i \le M-1 \\ \pi_M &= \left(\frac{p}{q}\right)^{M-1}\pi_0 \end{aligned}$$

and finally

$$\lim_{n\to\infty} p_{ii}(2n) = 2\pi_i$$

Some special cases:

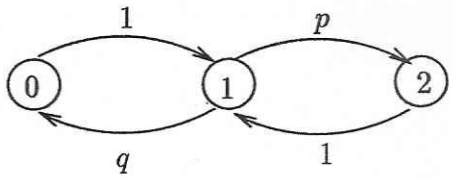
$M = 1$



$$\pi_0 = \frac{1}{2} \quad \Rightarrow \quad p_{00}(2n) = 1$$
$$p_{11}(2n) = 1$$

that is, an even number of steps implies certain return to the same state.

$M = 2$



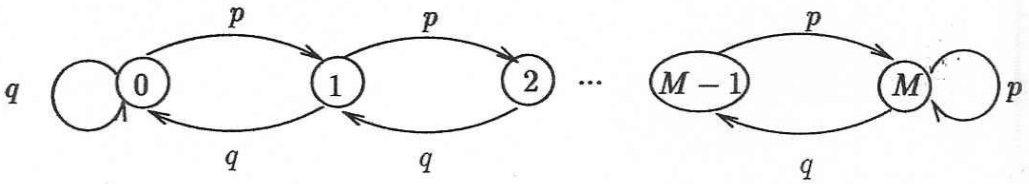
$$\pi_0 = \frac{q}{2} \qquad \pi_1 = \frac{1}{2} \qquad \pi_2 = \frac{p}{2}$$

Note $p_{11}(2n) = 1$, that is, every other step involves a return to state 1.

If $p = q = \frac{1}{2}$

$$1 = \pi_0(1 + \underbrace{2 + 2 + \ldots + 2}_{M-1} + 1) = 2M\pi_0$$
$$\Rightarrow \pi_0 = \pi_M = \frac{1}{2M} \qquad \pi_i = \frac{1}{M} \qquad 2 \leq i \leq M-1$$

11.31



This is an aperiodic Markov chain $\Rightarrow p_{ii}(n) \to \pi_i$ as $n \to \infty$

$$\pi_0 = q\pi_0 + q\pi_1 \quad \Rightarrow \quad \pi_1 = \frac{p}{q}\pi_0$$

$$\pi_1 = p\pi_0 + q\pi_2 \qquad \pi_2 = \frac{1}{q}(\pi_1 - p\pi_0) = \frac{1}{q}\left(\frac{p}{q} - p\right)\pi_0 = \left(\frac{p}{q}\right)^2 \pi_0$$

$$\pi_2 = p\pi_1 + q\pi_3 \qquad \pi_3 = \frac{1}{q}(\pi_2 - p\pi_1) = \frac{1}{q}\left(\frac{p^2}{q^2} - \frac{p^2}{q}\right)\pi_0$$

$$\vdots \qquad \frac{1}{q}\frac{p^2}{q}\left(\frac{1}{q} - 1\right) = \frac{p^3}{q^3}\pi_0$$

$$\pi_{M-1} = p\pi_{M-1} + q\pi_M \qquad \vdots$$

$$\pi_M = p\pi_M + p\pi_{M-1} \qquad \pi_M = \left(\frac{p}{q}\right)^M \pi_0$$

$$1 = \pi_0\left(1 + \frac{p}{q} + \left(\frac{p}{q}\right)^2 + \ldots + \left(\frac{p}{q}\right)^M\right) = \pi_0 \frac{1 - \left(\frac{p}{q}\right)^{M+1}}{1 - \frac{p}{q}}$$

$$\pi_0 = \frac{1 - \frac{p}{q}}{1 - \left(\frac{p}{q}\right)^{M+1}}$$

$$\pi_i = \frac{1 - \frac{p}{q}}{1 - \left(\frac{p}{q}\right)^{M+1}}\left(\frac{p}{q}\right)^i$$

11.32

If an irreducible class has n states, then every state can be reached from every other state in n or fewer steps. If a state has zero probability then all states that lead to it must also have zero probability from the transition probability equations. But this implies that all the states must have zero probability, which is not possible for a finite state MC.

11.34

suppose i is a positive recurrent state.

If state i communicates with j, then there exist n, m such that $P_{ij}(n) > 0$, $P_{ji}(m) > 0$.

For any k one can see $P_{jj}(m+n+k) \geq P_{ji}(m)\, P_{ii}(n)\, P_{ij}(k)$

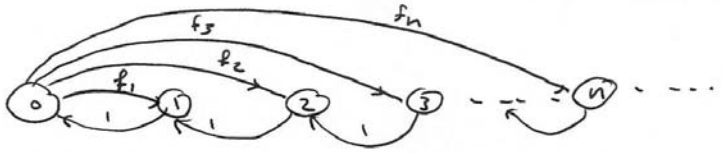
Since $P_{jj}(m+n+k)$ is the probability of going from j to j in $(m+n+k)$ steps, while $P_{ji}(m)\, P_{ii}(n)\, P_{ij}(k)$ is the probability of going from j to i in m steps and then from i to i in (n) steps and finally from i to j in k steps.

$$\Rightarrow \sum_{n=1}^{\infty} P_{jj}(m+n+k) \geq P_{ji}(m)\, P_{ij}(k) \underbrace{\sum_{n=1}^{\infty} P_{ii}}_{\infty \ (i \text{ is pos. recurrent})} = \infty$$

$\Rightarrow j$ is also positive recurrent

11.36

(a)



(b) There is a single class in this Markov chain $\Rightarrow$ irreducible

(c) state zero is recurrent

(d) $x = (x_0, x_1, \cdots x_n \cdots)$

$\Rightarrow \quad xP = x \quad \Rightarrow$

$x_1 = x_0$

$f_1 x_0 + x_2 = x_1 \Rightarrow x_2 = (1-f_1)x_0$

$f_2 x_0 + x_3 = x_2 \Rightarrow x_3 = (1-f_1-f_2)x_0$

$P =$

	0	1	2	3	⋯
0	f_1	f_2	f_3	⋯	
1	0	0	0	—	
2	0	1	0	0	⋯
⋮	⋮		⋱	⋱	⋱

$\Rightarrow \quad \boxed{x_n = (1-f_1-f_2-\cdots-f_{n-1})x_0}$

$\sum_{n=0}^{\infty} x_n = 1 \quad \Rightarrow$

1
$1-f_1$
$1-f_1-f_2$
$1-f_1-f_2-f_3$
$\vdots$

$$\sum_{i=0}^{n-1} x_i = 1 x_0 + (1-f_1)x_0 + (1-f_1-f_2)x_0 + (1-f_1-f_2-f_3)x_0 + \cdots + (1-f_1-\cdots-f_{n-1})x_0$$

$$= \left[n - (n-1)f_1 - (n-2)f_2 - \cdots - f_{n-1}\right] x_0$$

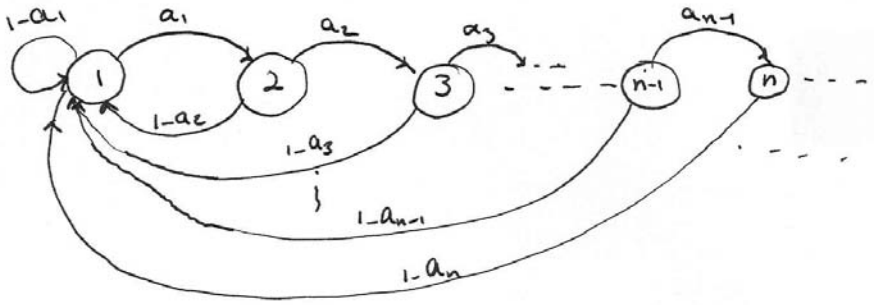
$$= \underbrace{\left[n(1-f_1-f_2-\cdots-f_{n-1}) + f_1 + 2f_2 + \cdots + (n-1)f_{n-1}\right]}_{K} x_0$$

$$\Rightarrow x_0 = \frac{1}{K} \qquad x_n = \frac{(1-f_1-f_2-\cdots-f_{n-1})}{K}$$

11.37 S = state space = $\{1, 2, 3, \cdots\}$

$\forall j \in S \quad P_{j,j+1} = a_j \; ; \quad P_{j1} = 1 - a_j \quad 0 < a_j < 1$

(a)



(b) All the states are reachable and construct one class

$\Rightarrow$ irreducible

(c) State 1 is positive recurrent.

(d)

$$P = \begin{bmatrix} 1-a_1 & a_1 & 0 & 0 & \cdots \\ 1-a_2 & 0 & a_2 & & \\ 1-a_3 & 0 & 0 & a_3 & \\ \vdots & & & & \ddots \end{bmatrix}$$

$\pi P = \pi$

$(1-a_1)\pi_1 + (1-a_2)\pi_2 + \cdots = \pi_1$

$a_1\pi_1 = \pi_2 \qquad \pi_2 = a_1\pi_1$

$a_2\pi_2 = \pi_3 \Rightarrow \quad \pi_3 = a_1a_2\pi_1$

$\vdots \qquad\qquad \vdots$

$\pi_{n+1} = a_1a_2\cdots a_n\pi_1$

$$\sum_{i=1}^{\infty} \pi_i = 1 \Rightarrow \pi_1 + a_1\pi_1 + a_2a_1\pi_1 + \cdots = 1$$

$$\pi_1 = \underbrace{\frac{1}{1 + a_1 + a_1a_2 + a_1a_2a_3 + \cdots}}_{\text{assume } A} = \frac{1}{A}$$

$$\begin{cases} \pi_1 = \frac{1}{A} \\ \pi_2 = \frac{a_1}{A} \\ \pi_3 = \frac{a_1a_2}{A} \\ \vdots \end{cases}$$

(e) i) $A = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \cdots = \frac{1}{1-\frac{1}{2}} = 2 \Rightarrow \pi = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\right)$

11.37 (d) (i) $a_j = \frac{j-1}{j} \Rightarrow a_1 = 0 \Rightarrow \begin{cases} \pi_1 = 1 \\ \pi_i = 0 \ i>1 \end{cases}$ always in state 1

(ii) $a_j = \frac{1}{j}$ $\quad A = 1 + 1 + 1\times\frac{1}{2} + 1\times\frac{1}{2}\times\frac{1}{3} + \cdots = e$

$\Rightarrow \pi_1 = \frac{1}{e} \qquad \pi_j = \frac{a_j}{A} = \frac{1}{j \times e}$

(iii) $a_j = (1/2)^j \Rightarrow A = 1.64$

$\pi_1 = \frac{1}{1.64} = 0.6098 \qquad \pi_2 = (\frac{1}{2}) \times 0.6098 = 0.3046 \cdots \quad \pi_{n+1} = (\frac{1}{2})^n \times 0.6098$

(iv) $a_j = 1 - (\frac{1}{2})^j \Rightarrow A =$

11.38

(a) suppose event A is $(X_{n+1}=j \mid X_n=i, X_{n-1}=i, \ldots)$

Then $P(A) = P(A|H)P(H) + P(A|T)P(T) = \frac{1}{2}P(A|H) + \frac{1}{2}P(A|T)$

but $\begin{cases} P(A|H) = (P_1)_{ij} \\ P(A|T) = (P_2)_{ij} \end{cases} \Rightarrow P(A) = \frac{1}{2}(P_1)_{ij} + \frac{1}{2}(P_2)_{ij}$

$\Rightarrow P = \frac{1}{2}(P_1+P_2) \Rightarrow$ Markov ergodic it has pmf

(b) This process is not Markov since at each time instant one needs to know the previous steps in order to be able to find the probability of X_{n+1}.

(c) In any time instant, if it is an even number P_2 is used and in case of an even number P_1.
In this case in fact in any time constant the transition Probability matrix alternates between P_1 & P_2.
If π_0 be the initial state, then the other states can be calculated:

$$\pi_1 = \pi_0 P_1$$
$$\pi_2 = \pi_1 P_2 = \pi_0(P_1 P_2)$$
$$\pi_3 = \pi_0(P_1 P_2 P_1)$$
$$\pi_4 = \pi_0(P_1 P_2)^2$$
$$\begin{cases} \pi_{2n} = \pi_0(P_1 P_2)^n \\ \pi_{2n+1} = \pi_0(P_1 P_2)^n P_1 \end{cases} \Rightarrow$$ we have two recurrent classes (two pmf exists)

11.4 Continuous-Time Markov Chains

11.41

~~8.14~~ From Ex.11.36 we have

$$p_0(t) = \frac{\beta}{\alpha+\beta} + \left(p_0(0) - \frac{\beta}{\alpha+\beta}\right) e^{-(\alpha+\beta)t}$$
$$p_1(t) = \frac{\alpha}{\alpha+\beta} + \left(p_1(0) - \frac{\alpha}{\alpha+\beta}\right) e^{-(\alpha+\beta)t}$$

a) Now suppose we know the initial state is 0, then $p_0(0) = 1 \Rightarrow$

$$p_{00}(t) = \frac{\beta}{\alpha+\beta} + \left(1 - \frac{\beta}{\alpha+\beta}\right) e^{-(\alpha+\beta)t} = \frac{\beta + \alpha e^{-(\alpha+\beta)t}}{\alpha+\beta}$$
$$p_{01}(t) = 1 - p_{00}(t) = \frac{\alpha(1 - e^{-(\alpha+\beta)t})}{\alpha+\beta}$$

If the initial state is 1, then $p_1(0) = 1 \Rightarrow$

$$p_{11}(t) = \frac{\alpha}{\alpha+\beta} + \left(1 + \frac{\alpha}{\alpha+\beta}\right) e^{-(\alpha+\beta)t} = \frac{\alpha + \beta e^{-(\alpha+\beta)t}}{\alpha+\beta}$$
$$p_{10}(t) = 1 - p_{11}(t) = \frac{\beta(1 - e^{-(\alpha+\beta)t}}{\alpha+\beta}$$
$$\therefore \mathbb{P}(t) = \frac{1}{\alpha+\beta} \begin{bmatrix} \beta + \alpha e^{-(\alpha+\beta)t} & \alpha(1 - e^{-(\alpha+\beta)t}) \\ \beta(q - e^{-(\alpha+\beta)t}) & \alpha + \beta e^{-(\alpha+\beta)t} \end{bmatrix}$$

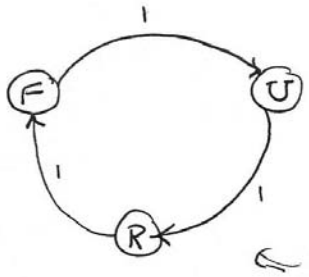
b) $P[X(1.5) = 1, X(3) = 1/X(0) = 0]$

$$= P[X(3) = 1/X(1.5) = 1, X(0) = 0]P[X(1.5) = 1/X(0) = 0]$$
$$= P[X(3) = 1/X(1.5) = 1]P[X(1.5) = 1/X(0) = 0]$$
$$= p_{11}(1.5)p_{01}(1.5)$$

$$P[X(1.5) = 1, X(3) = 1] = P[X(3) = 1/X(1.5) = 1]P[X(1.5) = 1]$$
$$= p_{11}(1.5)\left[\frac{\alpha}{\alpha+\beta} + \left(p_1(0) - \frac{\alpha}{\alpha+\beta}\right) e^{-(\alpha+\beta)1.5}\right]$$

11.42

(a)



From Full-charged (F) one can go to "in use" (U) state and from in use to "recharge" (R) only from "R" one can go to "F"

⇐ embedded Markov chain

(b)

$E(T_i) = \frac{1}{\nu_i}$ $i = F, U, R$ $\Rightarrow \begin{cases} \nu_F = \lambda \\ \nu_U = 1 \\ \nu_R = \frac{1}{3} \end{cases}$

$\Pi_F = \Pi_U = \Pi_R = \frac{1}{3}$

(stationary probability of embedded Markov chain)

$$P_i = \frac{\frac{\Pi_i}{\nu_i}}{\sum_j \frac{\Pi_j}{\nu_j}}$$

$$P_F = \frac{\frac{1}{\lambda} \times \frac{1}{3}}{\frac{1}{3}\left(\frac{1}{\lambda}+1+3\right)} = \frac{1}{4\lambda+1}$$

$$P_U = \frac{1 \times \frac{1}{3}}{\frac{1}{3}\left(\frac{1}{\lambda}+1+3\right)} = \frac{\lambda}{4\lambda+1}$$

$$P_R = \frac{3 \times \frac{1}{3}}{\frac{1}{3}\left(\frac{1}{\lambda}+1+3\right)} = \frac{3\lambda}{4\lambda+1}$$

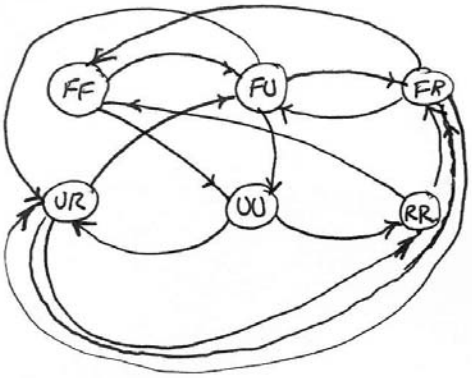
for $\lambda \to 0^+$ $P_F \to 1$ and $\begin{cases} P_U \to 0 \\ P_R \to 0 \end{cases}$ which is expected since $\lambda \to 0^+$ means that the occupancy time for state "F" is going to ∞.

for $\lambda \to +\infty$ $P_F \to 0$ and $\begin{cases} P \to \frac{1}{4} \\ P \to \frac{3}{4} \end{cases}$

Again, when $\lambda \to \infty$ the occupancy time of state "F" is negligible and the whole time is shared between "U" and "R" states

11.43

(a)



	FF	FU	FR	UR	UU	RR
FF	0	$\frac{2}{3}$	0	0	$\frac{1}{3}$	0
FU	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
FR	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$	0	0
UR	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	$\frac{1}{3}$
UU	0	0	0	$\frac{2}{3}$	0	$\frac{1}{3}$
RR	$\frac{1}{3}$	0	$\frac{2}{3}$	0	0	0

$\Rightarrow \Pi = \left(\frac{1}{9} \quad \frac{2}{9} \quad \frac{2}{9} \quad \frac{2}{9} \quad \frac{1}{9} \quad \frac{1}{9}\right)$ by steady state of embedded Markov chain

(b) To find P_i we notice that:

$$P(T_m > t \ \& \ T_n > t) = P(T_m > t) \times P(T_n > t) = e^{-(v_m + v_n)t}$$

$$\Rightarrow \quad v_{mn} = v_m + v_n \Rightarrow$$

$v_{FF} = 2\lambda$; $v_{RR} = \frac{2}{3}$ $v_{UU} = 2$

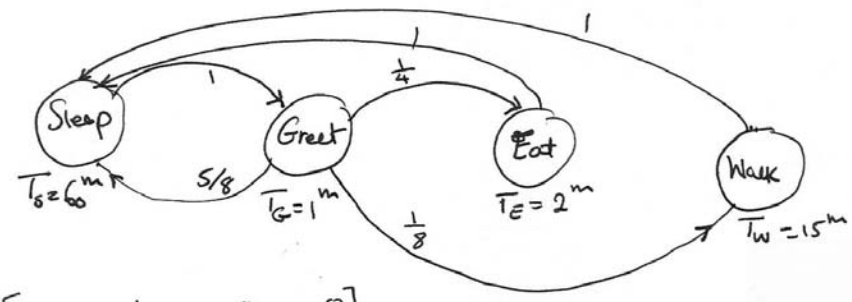
$v_{FR} = \lambda + \frac{1}{3}$; $v_{FU} = \lambda + 1$ $v_{UR} = 1 + \frac{1}{3} = \frac{4}{3}$

$$P_i = \frac{\frac{\pi_i}{v_i}}{\sum \frac{\pi_i}{v_i}} \Rightarrow P_{FF} = \frac{\frac{1}{9} \times \frac{1}{2\lambda}}{\frac{1}{2\lambda} + \frac{3}{3\lambda+1} + \frac{1}{\lambda+1} + \frac{11}{4}}$$

The rest can be obtained in a similar way.

When $\lambda \to \infty$ $P_{FF} \to 0$ and when $\lambda \to 0^+$ $P_{FF} \to \frac{1}{9}$

11.44 (a)



$$M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 5/8 & 0 & 1/4 & 1/8 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$\pi M = \pi \Rightarrow$

$$\begin{cases} \frac{5}{8}\pi_1 + \pi_2 + \pi_3 = \pi_0 \\ \pi_0 = \pi_1 \\ \frac{1}{4}\pi_1 = \pi_2 \\ \frac{1}{8}\pi_1 = \pi_3 \end{cases} \qquad \begin{array}{l} \pi_1 = \pi_0 \\ \pi_2 = \frac{1}{4}\pi_0 \\ \pi_3 = \frac{1}{8}\pi_0 \end{array}$$

$\Rightarrow \pi = \pi_0 (1, 1, \frac{1}{4}, \frac{1}{8})$

Normalization

$\pi = (\frac{8}{19}, \frac{8}{19}, \frac{2}{19}, \frac{1}{19})$

$\nu_S = \frac{1}{60}$, $\nu_G = \frac{1}{1}$, $\nu_E = \frac{1}{2}$, $\nu_W = \frac{1}{15}$

$$P_i = \frac{\frac{\pi_i}{\nu_i}}{\sum \frac{\pi_i}{\nu_i}} \Rightarrow P_S = \frac{\frac{8}{19} \times 60}{\frac{8}{19}\times 60 + \frac{8}{19}\times 1 + \frac{2}{19}\times 2 + \frac{1}{19}\times 15} = \frac{480}{507} = \frac{160}{169}$$

$$P_G = \frac{8}{507} \qquad P_E = \frac{4}{507} \qquad P_W = \frac{15}{507}$$

11.44

~~8.15~~ Let $N(t) = \#$ of spares at time t

$N(t)$ decreases by one each time a part breaks down, and the time between breakdowns is independent exponential RV's with rate α.

a)

$$\begin{aligned} p_{ij}(t) &= P[N(s+t) = j | N(s) = i] \quad 1 \le j \le i \le n \\ &= P[i - j \text{ breakdowns in time } t] \\ &= \frac{(\alpha t)^{i-j}}{(i-j)!} e^{-\alpha t} \\ p_{i0}(t) &= P[i \text{ or more "breakdowns" in time } t] \\ &= 1 - \sum_{k=0}^{i-1} \frac{(\alpha t)^k}{k!} e^{-\alpha t} \end{aligned}$$

b) $$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 1-e^{-\alpha t} & e^{-\alpha t} & 0 & 0 & \cdots & \\ 1 - \sum_{k=0}^{1} \frac{(\alpha t)^k}{k!} e^{-\alpha t} & \alpha t e^{-\alpha t} & e^{-\alpha t} & 0 & & \vdots \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 - \sum_{k=0}^{n-1} \frac{(\alpha t)^k}{k!} e^{-\alpha t} & \frac{(\alpha t)^{n-1}}{(n-1)!} e^{-\alpha t} & & \cdots & \alpha t e^{-\alpha t} & e^{-\alpha t} \end{bmatrix}$$

c) $p_n(0) = 1$

$$\begin{aligned} \Rightarrow p_j(t) &= p_{nj}(t) \qquad 1 \le j \le n \\ p_0(t) &= 1 - \sum_{j=0}^{n-1} p_j(t) \end{aligned}$$

11.45

8.16 a) For $j \leq i \leq n$

$$\begin{aligned} p_{ij}(t) &= P[N(s+t) = j | N(s) = i] \\ &= P[i-j \text{ machines out of } i \text{ machines break down} \\ &\quad \text{in } t \text{ seconds}] \\ &= \binom{i}{i-j} p^{i-j}(1-p)^j \end{aligned}$$

where

$$\begin{aligned} p &= P[\text{a particular machine breaks down by } t \text{ seconds}] \\ &= 1 - e^{-\alpha t} \end{aligned}$$

$$\therefore p_{ij}(t) = \binom{i}{i-j} (1-e^{-\alpha t})^{i-j} (e^{-\alpha t})^j$$

b)
$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1-e^{-\alpha t} & e^{-\alpha t} & 0 & \cdots & 0 \\ (1-e^{-\alpha t})^2 & 2(1-e^{-\alpha t})e^{-\alpha t} & e^{-2\alpha t} & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ (1-e^{-\alpha t})^n & n(1-e^{-\alpha t})^{n-1}e^{-\alpha t} & \cdots & & e^{-n\alpha t} \end{bmatrix}$$

c) $p_n(0) = 1$

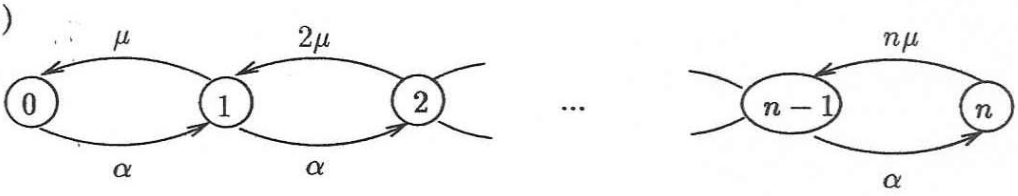
$$\Rightarrow p_j(t) = p_{nj}(t) \qquad 0 \leq j \leq n$$

11.46

8.17 Let $\tau_i \triangleq$ time till next breakdown of machine i

a)
$$\begin{aligned} T &\triangleq \text{time till next breakdown of any machine} \\ \Rightarrow T &= \min(\tau_1, \tau_2, ..., \tau_k) \\ P[T > t] &= P[\min(\tau_1, ..., \tau_k) > t] \\ &= P[\tau_1 > t, \tau_2 > t, ..., \tau_k > t] \\ &= P[\tau_1 > t]P[\tau_2 > t]...P[\tau_k > t] \\ &= (e^{-\mu t})^k = e^{-k\mu t} \end{aligned}$$

b)



$$[\gamma_{ij}] = \begin{bmatrix} -\alpha & \alpha & 0 & 0 & \dots & & \\ \mu & -(\alpha+\mu) & \alpha & 0 & \dots & & \\ \vdots & & & \ddots & & & \\ & & & (n-1)\mu & -(\alpha+(n-1))\mu & \alpha \\ 0 & 0 & \dots & 0 & n\mu & -n\mu \end{bmatrix}$$

c)
$$\begin{aligned} (\alpha + j\mu)P_j &= \alpha P_{j-1} + (n+1)\mu P_{j+1} \qquad 0 < j \leq n-1 \\ \alpha P_0 &= \mu P_1 \end{aligned}$$

Following M/M/1 example,

$$\begin{aligned} j\mu P_j &= \alpha P_{j-1} \\ \Rightarrow P_j &= \frac{\alpha}{j\mu} P_{j-1} \\ \Rightarrow P_j &= \frac{(\alpha/\mu)^j}{j!} P_0 \end{aligned}$$

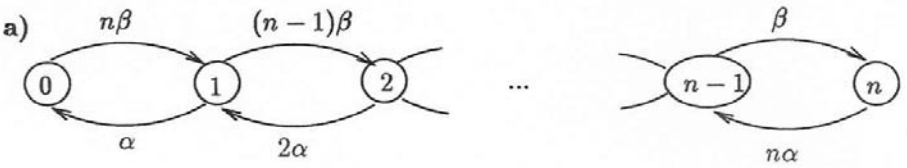
where

$$1 = \sum_{j=0}^{n} \frac{(\alpha/\mu)^j}{j!} P_0 \qquad P_0 = \frac{1}{\sum_{j=0}^{n} \frac{(\alpha/\mu)^j}{j!}}$$

11.47

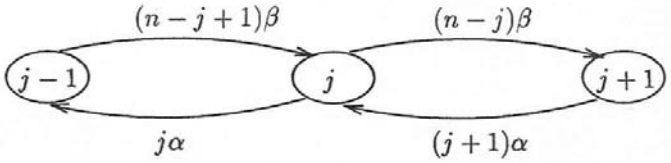
~~8.18~~ Suppose $N(t) = k$ speakers are active, then from Problem 11.46a the time until the next speaker goes silent is an exponential RV with rate $k\alpha$ and the transition rate from state k to $k-1$ is $k\alpha$. When $N(t) = k$, $n-k$ speakers are silent, and the time till the next speaker goes active is an exponential RV with rate $(n-k)\beta$; thus the transition rate from k to $k+1$ is $(n-k)\beta$.

a)



$$[\gamma_{ij}] = \begin{bmatrix} -n\beta & n\beta & 0 & 0 & \cdots & & 0 \\ \alpha & -(\alpha+(n-1)\beta) & (n-1)\beta & 0 & \cdots & & 0 \\ \alpha & -(\alpha+(n-1)\beta) & (n-1)\beta & 0 & \cdots & & 0 \\ 0 & 2\alpha & -(2\alpha+(n-2)\beta) & (n-2)\beta & \cdots & & 0 \\ \vdots & & & \ddots & \ddots & & \\ & & & & (n-1)\alpha & -(n-1)\alpha-\beta & \beta \\ 0 & & \cdots & & 0 & n\alpha & -n\alpha \end{bmatrix}$$

b)



$$\begin{aligned} [j\alpha + (n-j)\beta]P_j &= (n-j+1)\beta P_{j-1} + (j+1)\alpha P_{j+1} \quad 0 < j < n-1 \\ n\beta P_0 &= \alpha P_1 \end{aligned}$$

Proceeding as in M/M/1 example, we can show that the above are equivalent to

$$\begin{aligned} (n-j)\beta P_j &= (j+1)\alpha P_{j+1} \\ \Rightarrow P_{j+1} &= \frac{n-j}{j+1}\left(\frac{\beta}{\alpha}\right)P_j \quad (\star)\ j = 0, \ldots, n-1 \end{aligned}$$

Claim: $P_j = \binom{n}{j}\left(\frac{\beta}{\alpha}\right)^j P_0$

Proof: for $j = 0 \qquad P_0 = P_0 \checkmark$
Assume P_j as above, then $(\star)$ implies

$$\begin{aligned} P_{j+1} &= \frac{n-j}{j+1}\left(\frac{\beta}{\alpha}\right)\frac{n!}{(n-j)!j!}\left(\frac{\beta}{\alpha}\right)^j P_0 \\ &= \frac{n!}{(n-j-1)!(j+1)!}\left(\frac{\beta}{\alpha}\right)^{j+1} P_0 \\ &= \binom{n}{j+1}\left(\frac{\beta}{\alpha}\right)^{j+1} P_0 \quad \checkmark \\ 1 &= P_0 \sum_{j=1}^{n} \binom{n}{j}\left(\frac{\beta}{\alpha}\right)^j = P_0\left(\frac{\beta}{\alpha}+1\right)^n \\ &\Rightarrow P_0 = \left(\frac{\alpha}{\alpha+\beta}\right)^n \\ \Rightarrow P_j &= \binom{n}{j}\left(\frac{\beta}{\alpha}\right)^j\left(\frac{\alpha}{\alpha+\beta}\right)^n = \binom{n}{j}\left(\frac{\beta}{\alpha+\beta}\right)^j\left(\frac{\alpha}{\alpha+\beta}\right)^{n-j} \qquad 0 \le j \le n \end{aligned}$$

This is a binomial distribution for n independent speakers where each speaker is active with probability $\frac{\beta}{\alpha+\beta}$.

11.48

(a) Assuming $\delta \ll 1$ we can recognize three situations in each state. One arrival can come with the probability of $\lambda\delta$. One departure can occur with prob. $\mu\delta$, and no-arrival or departure with prob. $1-(\mu+\lambda)\delta$.

The Transition prob. matrix can be written as:

$P_{j,j+1} = \lambda\delta \qquad P_{j,j-1} = \delta\mu \qquad P_{j,j} = 1-(\lambda+\mu)\delta \qquad P_{00} = (1-\lambda)\delta \qquad P_{01} = \lambda\delta$

$$P = \begin{pmatrix} (1-\lambda)\delta & \delta\lambda & 0 & 0 & \cdots & \cdots & \cdots \\ \delta\mu & 1-(\lambda+\mu)\delta & \delta\lambda & 0 & 0 & \cdots & \\ 0 & \delta\mu & 1-(\lambda+\mu)\delta & \delta\lambda & 0 & \cdots & \\ 0 & 0 & \delta\mu & 1-(\lambda+\mu)\delta & \delta\lambda & & \\ \vdots & \vdots & & \ddots & \ddots & \ddots & \\ \vdots & \vdots & & \cdots & \cdots & \cdots & \end{pmatrix}$$

(b) $\pi = \pi P \qquad \pi = (\pi_0, \pi_1, \cdots)$

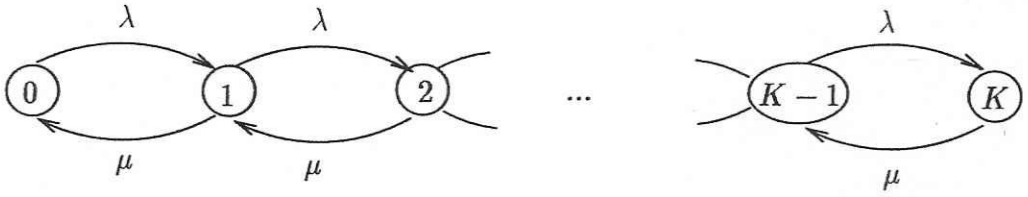
$$\Rightarrow \begin{cases} (1-\lambda)\delta\,\pi_0 + \delta\mu\,\pi_1 = \pi_0 \\ \delta\lambda\,\pi_0 + [1-(\lambda+\mu)\delta]\pi_1 + \delta\mu\pi_2 = \pi_1 \\ \vdots \\ \delta\lambda\,\pi_{i-1} + [1-(\lambda+\mu)\delta]\pi_i + \delta\mu\,\pi_{i+1} = \pi_i \end{cases} \Rightarrow \pi_1 = \frac{\lambda}{\mu}\pi_0$$

$$\Rightarrow \delta\lambda\,\pi_{i-1} + \delta\mu\,\pi_{i+1} = (\lambda+\mu)\delta\,\pi_i$$

$$\begin{cases} \pi_1 = \frac{\lambda}{\mu}\pi_0 = \rho\,\pi_0 \\ \lambda\pi_{i-1} + \mu\,\pi_{i+1} = (\lambda+\mu)\pi_i \end{cases} \Rightarrow \boxed{\pi_i = \left(1-\frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^i = (1-\rho)(\rho)^i}$$

11.49

8.19 The transition rate diagram is:



Eqn.11.42 applies here, so we have

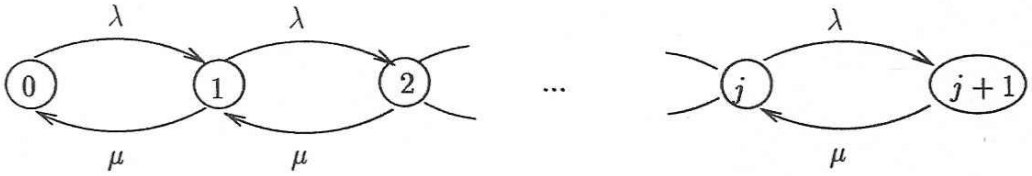
$$P_{j+1} = \left(\frac{\lambda}{\mu}\right) P_j = \left(\frac{\lambda}{\mu}\right)^{j+1} P_0$$

To find P_0 consider

$$\begin{aligned} 1 &= P_0 \sum_{j=0}^{K} \left(\frac{\lambda}{\mu}\right)^j = P_0 \frac{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}}{1 - \frac{\lambda}{\mu}} \\ \Rightarrow P_j &= \frac{\left(1 - \frac{\lambda}{\mu}\right)}{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}} \left(\frac{\lambda}{\mu}\right)^j \qquad 0 \le j \le K \end{aligned}$$

11.50

8.24 The transition rate diagram for the continuous-time process is:



Let

T_+ be exponential RV with parameter λ
T_- be exponential RV with parameter μ

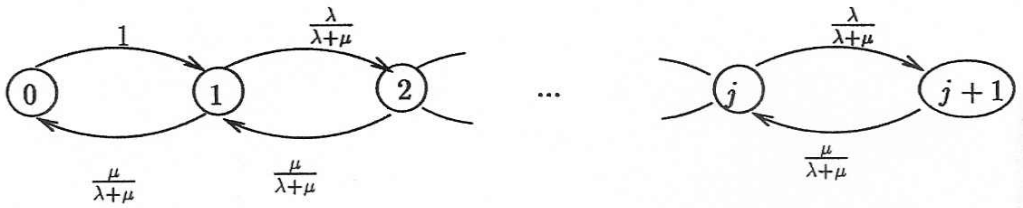
Then

$$P_{i,i+1} = P[T_+ < T_-] = 1 - P_{i,i-1}$$

From the solution of Problem 9.37 we know that

$$P[T_+ < T_-] = \frac{\lambda}{\lambda + \mu}$$

Thus transition probability diagram for the embedded process is:



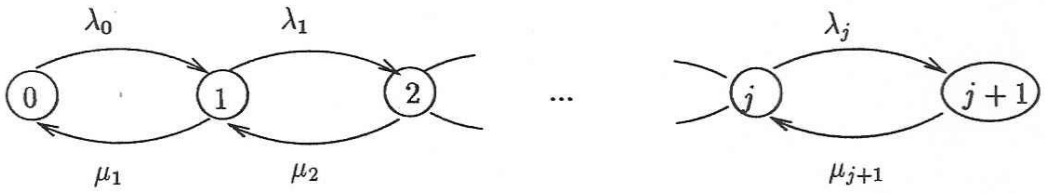
This discrete-time Markov chain is the one discussed in Problem 11.30 with $M = \infty$, thus letting $p = \frac{\lambda}{\lambda+\mu} < \frac{\lambda}{\lambda+\lambda} = \frac{1}{2}$

$$\pi_i = \frac{p^{i-1}}{q^i}\pi_0$$

and

$$\begin{aligned}
1 &= \pi_0\left[1 + \frac{1}{q}\underbrace{\left(1 + \frac{p}{q} + \frac{p}{q^2} + \ldots\right)}_{\frac{1}{1-\frac{p}{q}}}\right] \\
1 &= \pi_0\left[1 + \frac{1}{q-p}\right] = \pi_0\left[1 + \frac{1}{1-2p}\right] \\
&= \pi_0\frac{1-2p+1}{1-2p} \\
\pi_0 &= \frac{1-2p}{2(1-p)} = \frac{1-\frac{2\lambda}{\lambda+\mu}}{2\frac{\mu}{\lambda+\mu}} = \frac{\lambda+\mu-2\lambda}{2\mu} = \frac{\mu-3}{2\mu} \\
&= \frac{1-p}{2} \\
\pi_i &= \frac{1}{q}\left(\frac{p}{q}\right)^{i-1}\pi_0 = \frac{\lambda+\mu}{\mu}\left(\frac{\lambda}{\mu}\right)^{i-1}\frac{\mu-\lambda}{2\mu} \\
P_0 &= c\frac{\mu-\lambda}{2\mu}\frac{1}{\lambda} = c\frac{(1-\rho)}{2\lambda} \\
P_i &= c\frac{\lambda+\mu}{\mu}\left(\frac{\lambda}{\mu}\right)^{i-1}\left(\frac{1-\rho}{2\mu}\right)\frac{1}{\lambda+\mu} \\
&= c(1-\rho)\frac{1}{2\mu}\left(\frac{\lambda}{\mu}\right)^{i-1} \\
1 &= \underbrace{\frac{c(1-\rho)}{2}\left[\frac{1}{\lambda} + \frac{1}{\lambda}\underbrace{\left[1 + \rho + \rho^2 + \ldots\right]}_{\frac{1}{1-\rho}=\frac{\mu}{\mu-\lambda}}\right]}_{\frac{\mu-\lambda+\lambda}{\lambda(\mu-\lambda)}=\frac{\mu}{\lambda(\mu-\lambda)}=\frac{1}{\lambda}\left(\frac{1}{1-\rho}\right)} \\
1 &= \frac{c}{2\lambda} \Rightarrow c = 2\lambda \\
&\Rightarrow P_i = (1-\rho)\rho^i \quad \surd
\end{aligned}$$

11.51



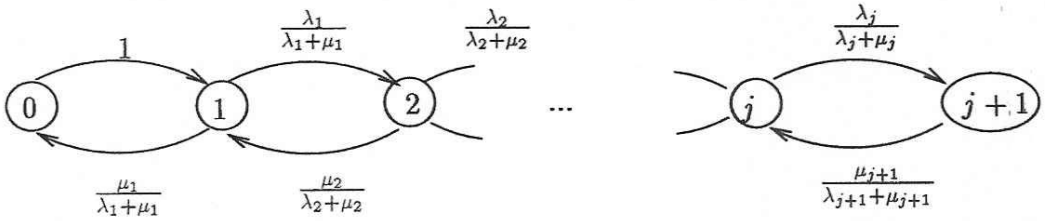
Let

$T_+(j)$ be exp RV with rate λ_j
$T_-(j)$ be exp RV with rate μ_j

then

$$P_{j,j+1} = P[T_+(j) < T_-(j)] = \frac{\lambda_j}{\lambda_j + \mu_j} = 1 - P_{j,j-1}$$

The transition probability diagram for the embedded draw is:



$$\begin{aligned}
\pi_0 &= \frac{\mu_1}{\lambda_1 + \mu_1}\pi_1 \\
\pi_1 &= \pi_0 + \frac{\mu_2}{\lambda_2 + \mu_2}\pi_2 \\
\pi_j &= \frac{\lambda_{j-1}}{\lambda_{j-1} + \mu_{j-1}}\pi_{j-1} + \frac{\mu_{j+1}}{\lambda_{j+1} + \mu_{j+1}}\pi_{j+1}
\end{aligned}$$

Substitute

$$\pi_j = \frac{P_j \nu_j}{c} = \frac{P_j(\lambda_j + \mu_j)}{c} \qquad j > 0 \qquad \pi_0 = \frac{P_0 \lambda_0}{c}$$

$$\left.\begin{aligned}
\lambda_0 P_0 &= \mu_1 P_1 \\
(\lambda_j + \mu_j)P_j &= \lambda_{j-1}P_{j-1} + \mu_{j+1}P_{j+1}
\end{aligned}\right\} \Rightarrow \text{global balance eqns.}$$

$\therefore$ solution to π_j eqns. consistent with P_j's found in Ex. 11.40.

11.52 a) oce. time is exp. with $\mu = 1$

$$\Rightarrow \nu_i = \frac{1}{E(T_i)} = 1$$

$$P_i = \frac{\pi_i/\nu_i}{\sum_j \pi_j/\nu_j} \quad \text{where from 11.36} \quad \pi_i = \frac{1-f_1-f_2-\cdots-f_{i-1}}{A}$$

$$A = 1 + 1 + (1-f_1) + (1-f_1-f_2) + (1-f_1-f_2-f_3) + \cdots)$$

$$\Rightarrow P_i = \frac{\pi_i}{\sum_j \pi_j} = \pi_i = \frac{1-\sum_{k=1}^{i-1} f_k}{A} = \frac{1-g_i}{A}$$

On the other hand f_i is the pmf of a geometric dist.

$$\text{then: } g_i = \sum_{k=1}^{i-1} f_k = \sum_{k=1}^{i-1} p(1-p)^{k-1} = p \times \frac{1-q^k}{1-q} = 1-q^k$$

$$\Rightarrow 1-g_i = q^k,$$

$$\text{and } A = 2 + \sum_{i=1}^{\infty}(1-g_i) = 2 + \sum_{i=1}^{\infty} q^i = 2 + \frac{q}{1-q} = \frac{2-q}{1-q}$$

$$\Rightarrow \boxed{P_i = \frac{1-g_i}{A} = q^i \frac{1-q}{2-q}}$$

b) In this part $\nu_i = \frac{1}{E(T_i)} = \frac{1}{i}$

$$P_i = \frac{\pi_i/i}{\sum_j \pi_j/j} = \frac{\frac{1}{i}q^i}{\sum_{j=1}^{\infty}\frac{1}{j}q^j} = \frac{\frac{1}{i}q^i}{-\ln(1-q)} = \frac{q^i}{i\ln(p)}$$

Since $\frac{1}{1-x} = 1+x+x^2+\cdots \Rightarrow \int \frac{dx}{1-x} = (x+\frac{1}{2}x^2+\frac{1}{3}x^3+\cdots)$

$$= -\ln(1-x)$$

$$\Rightarrow \sum_{j=1}^{\infty}\frac{1}{j}q^j = -\ln(1-q) = -\ln p$$

11.52 (c)

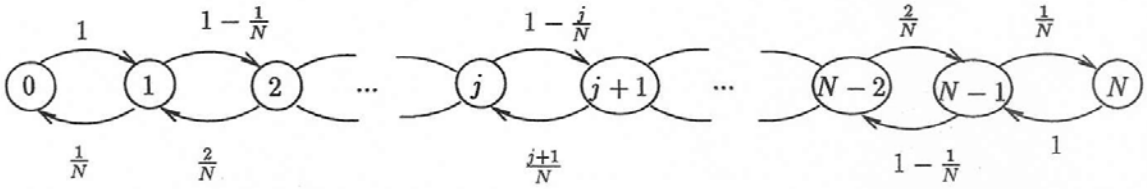
$$v_i = \frac{1}{E(T_i)} = 2^{-i}$$

$$P_i = \frac{\frac{\pi_i}{v_i}}{\sum_j \frac{\pi_j}{v_j}} = \frac{2^{-i}q^i}{\sum_{j=1}^{\infty} 2^{-j}q^j} = \frac{2^{-i}q^i}{\sum_{j=1}^{\infty} (q/2)^j} = \frac{2^{-i}q^i}{\frac{q/2}{1-q/2}}$$

$$\Rightarrow \boxed{P_i = \frac{2^{-i}q^{i-1}}{2-q}}$$

*11.5 Time-Reversed Markov Chains

11.53 a) The state transition diagram is:



This is a birth-death process, so by Example 11.44 it is reversible. If process is time reversible then

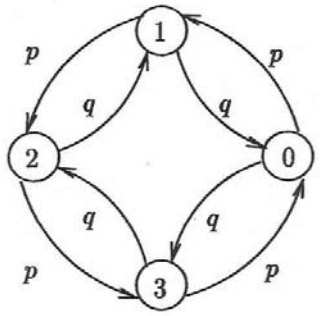
$$\begin{aligned} \pi_i p_{ij} &= \pi_j p_{ji} \\ \Rightarrow \pi_i p_{i,i+1} &= \pi_{i+1} p_{i+1,i} \\ \pi_i \left(1 - \frac{1}{N}\right) &= \pi_{i+1} \frac{i+1}{N} \\ \pi_{i+1} &= \frac{N-i}{i+1} \pi_i \end{aligned}$$

It is then easy to show that

$$\pi_j = \binom{N}{j} \pi_0$$

$$\begin{aligned} 1 = \pi_0 \sum_{j=0}^{N} \binom{N}{j} &= \pi_0 2^N \\ \Rightarrow \pi_j &= \binom{N}{j} \left(\frac{1}{2}\right)^N \end{aligned}$$

11.54



a)
$$\left.\begin{array}{rcl}\pi_0 &=& q\pi_1 + p\pi_3\\ \pi_1 &=& q\pi_2 + p\pi_0\\ \pi_2 &=& q\pi_3 + p\pi_1\\ \pi_3 &=& q\pi_0 + p\pi_2\end{array}\right\} \Rightarrow \pi_0 = \pi_1 = \pi_2 = \pi_3 = \frac{1}{4}$$

b) Clearly if $p > q$ then point tends to circulate counter clockwise whereas if $p < q$ it tends to circulate clockwise. Thus we expect that the process is time reversible only when $p = q = \frac{1}{2}$.

Indeed the condition for time reversibility requires that:

$$\pi_0 p_{01} = \pi_1 p_{10} \Leftrightarrow \underbrace{\pi_0 p}_{\frac{1}{4}} = \underbrace{\pi_1 q}_{\frac{1}{4}} \Leftrightarrow p = q$$

and similarly for the other states.

11.55

$$\begin{aligned} q_{ij} &= \frac{\pi_i p_{ji}}{\pi_i} = \frac{\left(\frac{p}{q}\right)^j p_{ji}}{\left(\frac{p}{q}\right)^i}\\ \Rightarrow q_{i,i+1} &= \frac{\left(\frac{p}{q}\right)^{i+1}}{\left(\frac{p}{q}\right)^i} p_{i+1,i} = \frac{p}{q} q = p = p_{i,i+1}\\ q_{i,i-1} &= \frac{\left(\frac{p}{q}\right)^{i-1}}{\left(\frac{p}{q}\right)^i} p_{i-1,i} = \frac{q}{p} p = q = p_{i,i-1}\\ q_{00} &= q = p_{00}\end{aligned}$$

$\Rightarrow$ Yes, process is time reversible.

11.56 From the solution of problem 11.16 we have

$$\pi_i = \frac{\binom{p}{i}^2}{\binom{2p}{p}} \quad \text{and} \quad \begin{cases} P_{i,i+1} = \dfrac{(p-i)^2}{p^2} \\ P_{i+1,i} = \dfrac{(i+1)^2}{p^2} \end{cases} \quad \text{and } P_{ij} = 0 \text{ for all } j \neq i \text{ or } j \neq i+1$$

one can see: $\pi_i P_{i,i+1} = \pi_i P_{ij} = \dfrac{\binom{p}{i}^2}{\binom{2p}{p}} \dfrac{(p-i)^2}{p^2}$ $\quad j = i+1$

and $\pi_j P_{ji} = \pi_{i+1} P_{i+1,i} = \dfrac{\binom{p}{i+1}^2}{\binom{2p}{p}} \times \dfrac{(i+1)^2}{p^2}$

$$= \frac{\frac{p!^2}{(i+1)!^2 (p-i-1)!^2}}{\binom{2p}{p}} \times \frac{(i+1)^2}{p^2}$$

$$= \frac{1}{\binom{2p}{p}} \times \frac{p!^2}{i!^2 (p-i)!^2} \times \frac{(p-i)^2}{(i+1)^2} \times \frac{(i+1)^2}{p^2}$$

$$= \frac{\binom{p}{i}^2}{\binom{2p}{p}} \times \frac{(p-i)^2}{p^2}$$

$$= \pi_i \times P_{ij}$$

$\Rightarrow \pi_j P_{ji} = \pi_i P_{ij}$ → time-reversible ✓

11.57 For the Markov chain of problem 11.17 we have:

$$P = \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \\ 0 & 0 & q & p \end{pmatrix} \Rightarrow \pi = (q^2, qp, qp, p^2)$$

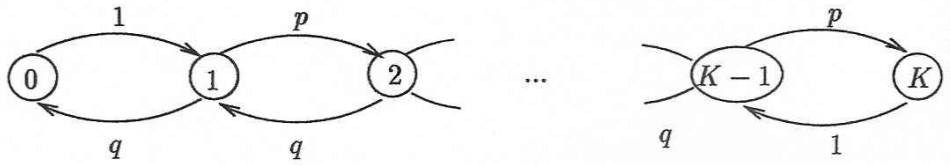
for these (i,j) pairs P_{ij} is non-zero $\{(0,1),(0,0)\}, \{(2,3),(2,2)\}, \{(3,1),(3,2)\}, \{(3,2),(3,3)\}$

where $\begin{cases} P_{01} = p \\ P_{10} = 0 \end{cases}$ $\begin{cases} \pi_0 P_{01} = q^2 \times p = q^2 p \\ \pi_1 P_{10} = qp \times 0 = 0 \end{cases} \Rightarrow \pi_0 P_{01} \neq P_1 P_{10}$

$\Rightarrow$ it is not time-reversible

11.58

8.29 a) The transition probability diagram for the embedded chain of the process in Problem 49 is:



where $p = \frac{\lambda}{\lambda+\mu}$, $q = \frac{\mu}{\lambda+\mu}$ (see Problem 11.50a).

In Problem 11.30 we showed that

$$x_0 = \frac{1-2p}{2\left(1-p\left(1+\left(\frac{p}{q}\right)^{K-1}\right)\right)}$$

$$\pi_i = \frac{1}{q}\left(\frac{p}{q}\right)^{i-1}\pi_0 \qquad 1 \le i \le K-1$$

$$\pi_K = \left(\frac{p}{q}\right)^{K-1}\pi_0$$

$\therefore$ the transition probabilities for the time reversed process are:

$$q_{01} = \frac{\pi_1}{\pi_0}\tilde{q}_{10} = \frac{1}{q}q = 1$$

$$q_{10} = \frac{\pi_0}{\pi_1}\tilde{q}_{01} = q \cdot 1 = q$$

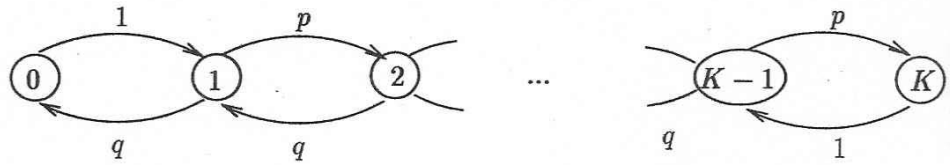
$$q_{i,i+1} = \frac{\pi_{i+1}}{\pi_i}\tilde{q}_{i+1,i} = \frac{p}{q}q = p$$

$$q_{i,i-1} = \frac{\pi_{i-1}}{\pi_1}\tilde{q}_{i-1,i} = \frac{q}{p}p = q$$

$$q_{K-1,K} = \frac{\pi_K}{\pi_{K-1}}\tilde{q}_{K,K-1} = p \cdot 1 = p$$

$$q_{K,K-1} = \frac{\pi_{K-1}}{\pi_K}\tilde{q}_{K-1,K} = \frac{1}{p}p = 1$$

$\Rightarrow$ transition probability diagram for reverse process is:



This is identical to forward process ⇒ process is reversible.

b) Eq. (11.67) implies

$$\begin{aligned} p_i\gamma_{i,i+1} &= p_{i+1}\gamma_{i+1,i} \\ \Rightarrow \lambda p_i &= \mu p_{i+1} \\ \Rightarrow p_{i+1} &= \frac{\lambda}{\mu}p_i \quad i \geq 0 \end{aligned}$$

which implies that $p_{i+1} = \left(\frac{\lambda}{\mu}\right)^{i+1} p_0$ and

$$p_i = \frac{1 = \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}} \left(\frac{\lambda}{\mu}\right)^i$$

as in Problem 11.49.

11.59 The Markov process in 11.37i is a time-reversed version of Ex 11.42 (Figure 11.15a) where $a_j = 1 - b_j$

From Ex 11.49 we have $\pi_j = \dfrac{P(X \geq j)}{E(X)}$

$$\pi_i P_{ij} = \pi'_j q_{ji} \Rightarrow \pi_i P_{i,i+1} = \pi'_{i+1} q_{i+1,i}$$

$$\frac{P(X \geq i)}{E(X)} \times (1 - b_i) = \pi'_{i+1} \times a_i \quad \left.\begin{array}{l} \\ 1 - b_i = a_i \end{array}\right\} \Rightarrow \pi'_{i+1} = \frac{P(X \geq i)}{E(X)}$$

$$\text{or} \quad \pi'_i = \frac{P(X \geq i-1)}{E(X)}$$

11.60 From Ex. 11.36 we have $\begin{cases} P_0 = \frac{\beta}{\alpha+\beta} & \gamma_{01} = \alpha \\ P_1 = \frac{\alpha}{\alpha+\beta} & \gamma_{10} = \beta \end{cases}$

$$\Rightarrow P_0 \gamma_{01} = \frac{\beta}{\alpha+\beta} \times \alpha = \frac{\alpha}{\alpha+\beta} \times \beta = P_1 \gamma_{10} \Rightarrow \text{reversible}$$

11.61 From problem 11.46 $\begin{cases} \gamma_{i,i+1} = \alpha \\ \gamma_{i+1,i} = (i+1)\mu \end{cases}$

$$P_i = \frac{(\alpha/\mu)^i}{i!} P_0$$

$$\Rightarrow P_i \gamma_{i,i+1} = \alpha \times \frac{(\frac{\alpha}{\mu})^i}{i!} P_0$$

$$P_{i+1} \gamma_{i+1,i} = \frac{(\alpha/\mu)^{i+1}}{(i+1)!} P_0 \times (i+1)\mu = \frac{\alpha(\frac{\alpha}{\mu})^i}{i!} P_0 = P_i P_{i,i+1}$$

$\Rightarrow$ reversible

11.62 From problem 11.47 $\begin{cases} \gamma_{i,i+1} = (n-i)\beta \\ \gamma_{i+1,i} = (i+1)\alpha \end{cases}$ $i \geq 3$ & $P_i = \binom{n}{i}(\frac{\beta}{\alpha})^i P_0$

$$\Rightarrow P_i \gamma_{i,i+1} = \binom{n}{i}(\tfrac{\beta}{\alpha})^i P_0 \times (n-i)\beta$$

$$P_{i+1} \gamma_{i+1,i} = \binom{n}{i+1}(\tfrac{\beta}{\alpha})^{i+1} P_0 \times (i+1)\alpha = \frac{n!}{(i+1)!(n-i-1)!}(\tfrac{\beta}{\alpha})^i P_0 (i+1)\beta$$

$$= \frac{n!\,(n-i)}{i!\,(n-i)!}(\tfrac{\beta}{\alpha})^i P_0 \beta = \binom{n}{i}(\tfrac{\beta}{\alpha})^i P_0 (n-i)\beta$$

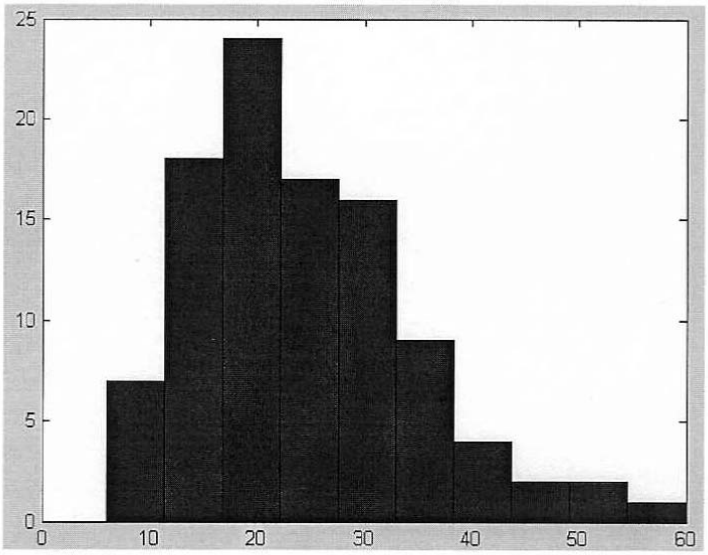
$$= P_i \gamma_{i,i+1}$$ reversible

11.6 Numerical Techniques for Markov Chains

11.63

```
n = 11;
b = 5;
w = 5;
iter=0;

for loop=1:100
   while (b > 0)
      f = floor(n*rand);
      if f<b
         b = b-1;
      end
      iter=iter+1;
   end
   res(loop)=iter;
   b=5;
   iter=0;
end
hist(res);
```



11.64

```
ro=20;

for i=1:200
   c1_w=floor((ro+1)*rand);
   c1_b=20-c1_w;
   c2_w=20-c1_w;
   c2_b=20-c2_w;

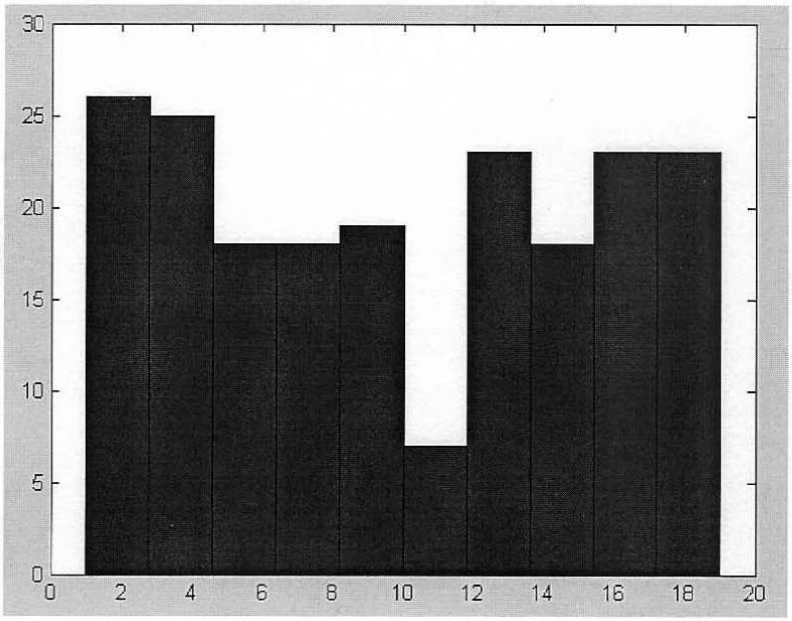
   f1=floor((ro+1)*rand);
   f2=floor((ro+1)*rand);

   if f1<=c1_w
      c2_w=c2_w+1;
      c1_w=c1_w-1;
   else
      c2_b=c2_b+1;
      c1_b=c1_b-1;
   end

   if f2<=c2_w
      c1_w=c1_w+1;
      c2_w=c2_w-1;
   else
      c1_b=c1_b+1;
      c2_b=c2_b-1;
   end

   res(i)=c1_w;
end

hist(res);
```



11.65

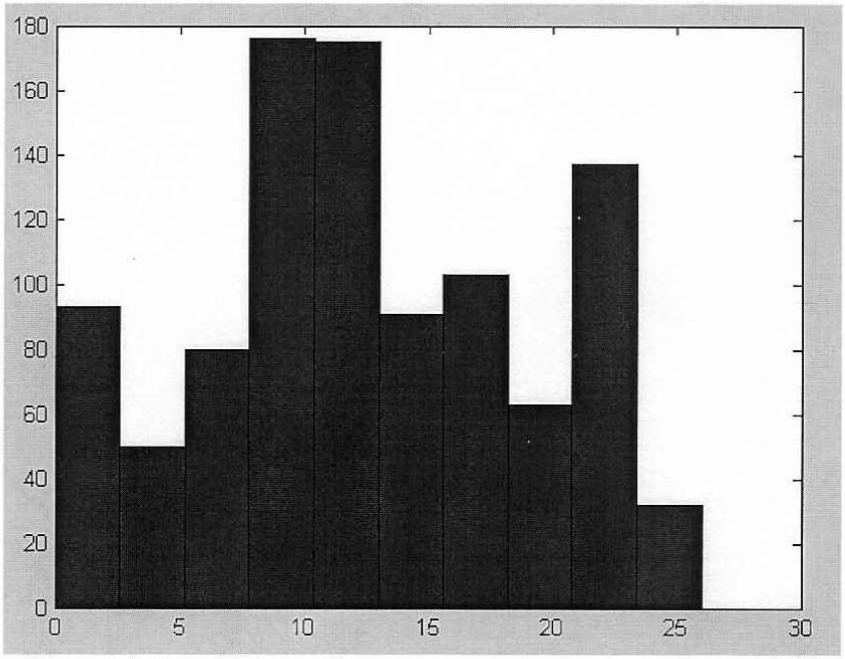
```
N=100;
b=0.5;
a=0.5;
iter=0;
X=0;

for i=1:1000
   p1=rand;
   if p1<a && X<N
      X=X+1;
   end

   p2=rand;
   if p2<b && X>0
      X=X-1;
   end

   res(i)=X;
end

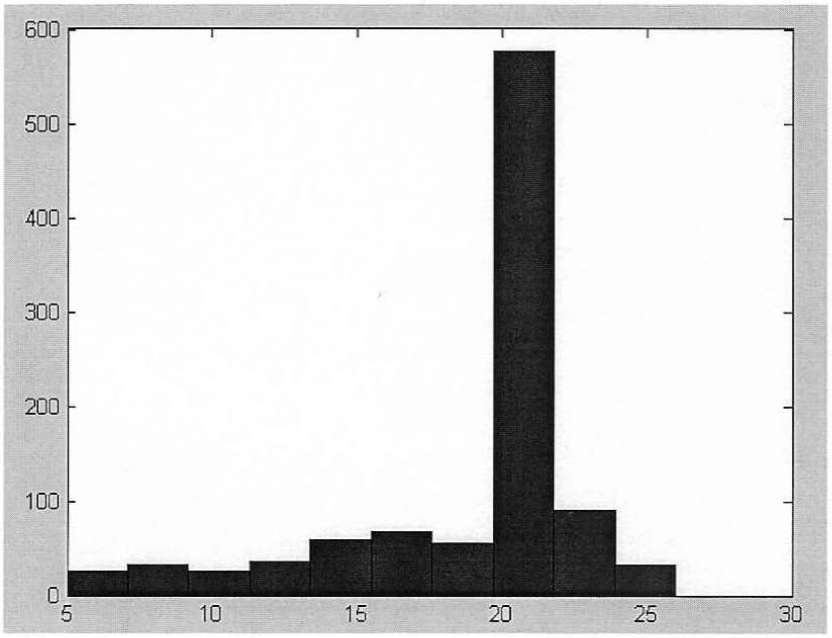
hist(res);
```



11.66

```
a=10;
b=5;

for i=1:500
while a>0 && b>0
      f=rand;
      if f<0.5
         a=a+1;
         b=b-1;
      else
         b=b+1;
         a=a-1;
      end
      iter=iter+1;
   end
   res(i)=iter;
end
hist(res);
```

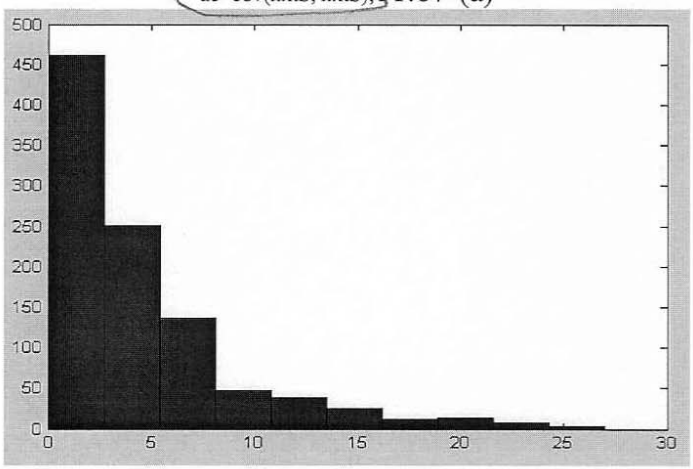


11.67

```
%For Geometric toggle=0, p=1/5. For Zipf toggle=1, p=5.
s=0;
N=1000;
toggle=0;
nxtS=zeros(N+1,1);
p=1/5;
for i=2:N+1
   if s==0
      if toggle==0
         s=randraw('geom',p,1);
      else
        s=randraw('zipf',p,1);
      end
      nxtS(i)=s;
   else
      s=nxtS(i-1)-1;
      nxtS(i)=s;
   end
end
hist(nxtS);
```

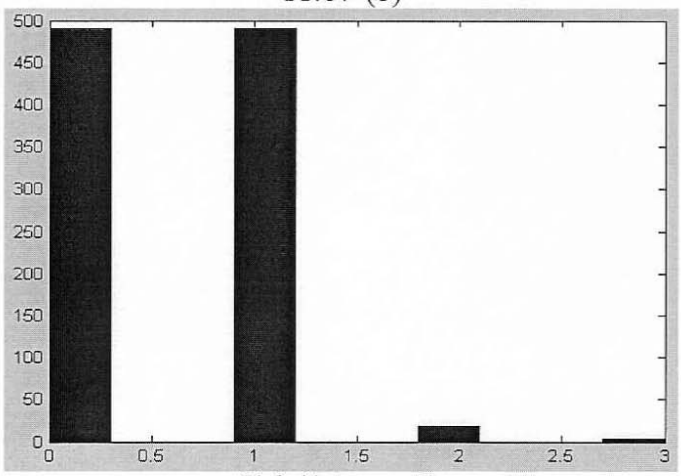
```
ac=cov(nxtS, nxtS);
```

11.67-(a)



Geometric with mean 5.

11.67-(b)



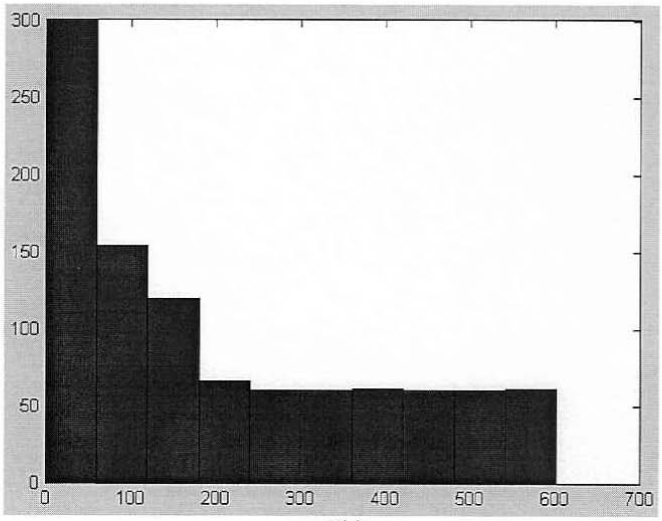
Zipf with parameter 5.

```
ac(geom) =[ 18.5959   18.5959        ac(zipf) =[  0.2759   0.2759
            18.5959   18.5959  ]                   0.2759   0.2759 ];
```

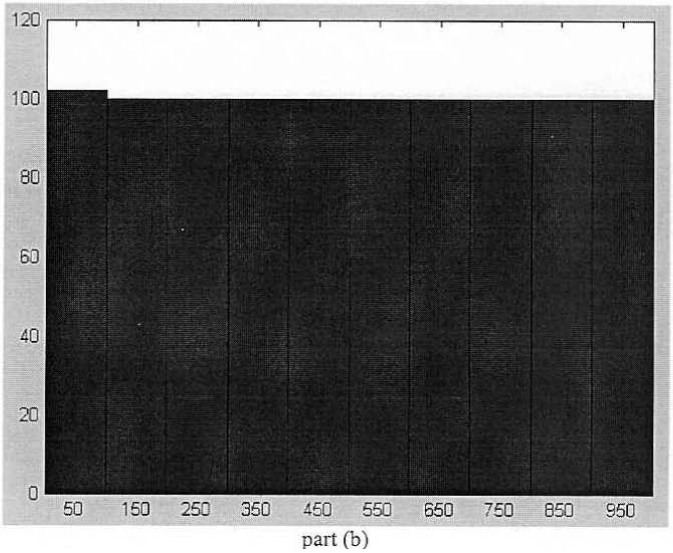
11.68

```
%For part (a) toggle=0. For part (2) toggle=1.
s=0;
a=0;
toggle=1;

for i=1:N+1
   if toggle==0
      a(i)=(i-1)/i;
   else
      a(i)=1-(1/2)^i;
   end
   if rand < a(i)
      s=s+1;
      nxtS(i+1)=s;
   else
      s=1;
      nxtS(i+1)=s;
   end
end
hist(nxtS);
```

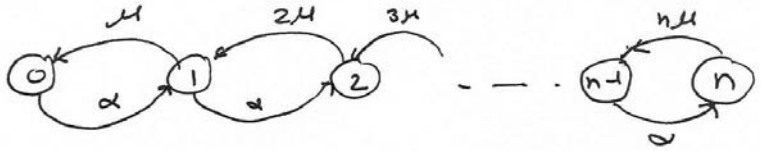


part (a)



part (b)

11.70



$$\Gamma = \begin{pmatrix} -\alpha & \alpha & 0 & 0 & \cdots & \\ \mu & -(\alpha+\mu) & \alpha & 0 & \cdots & \\ 0 & 2\mu & -(\alpha+2\mu) & \alpha & & \\ \vdots & \vdots & & \ddots & & \\ & & 0 & (n-1)\mu & & \alpha \\ 0 & & 0 & 0 & n\mu & -n\mu \end{pmatrix}$$

$n = 10$
$\mu = 0.1$
$\alpha = 1$

$$\Rightarrow \quad \Gamma = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0.1 & -1.1 & 1.0 & & & & \\ 0 & 0.2 & -1.2 & 1.0 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 0 & 0.9 & -1.9 & 1 \\ 0 & & & & 0 & 1 & -1 \end{pmatrix}$$

$$\begin{cases} (\alpha + i\mu) P_i = (n+1)\mu P_{i+1} + \alpha P_{i-1} & 0 \le i \le n-1 \\ \alpha P_0 = \mu P_1 \end{cases}$$

→ Recursive equations ∴ Program P_i

```
%P 1170
alpha=1;
mu=1/10;
n=10;

v=(alpha+mu);

L=[-alpha alpha 0 0 0 0 0 0 0 0 0
    mu -(alpha+mu) alpha 0 0 0 0 0 0 0 0
    0 2*mu -(alpha+2*mu) alpha 0 0 0 0 0 0 0
    0 0 3*mu -(alpha+3*mu) alpha 0 0 0 0 0 0
    0 0 0 4*mu -(alpha+4*mu) alpha 0 0 0 0 0
    0 0 0 0 5*mu -(alpha+5*mu) alpha 0 0 0 0
    0 0 0 0 0 6*mu -(alpha+6*mu) alpha 0 0 0
    0 0 0  0 0 0 7*mu -(alpha+7*mu) alpha 0 0
    0 0 0 0 0 0 0 8*mu -(alpha+8*mu) alpha 0
    0 0 0 0 0 0 0 0 9*mu -(alpha+9*mu) alpha
    0 0 0 0 0 0 0 0 0 10*mu -10*mu];

[E D]=eig(L);
t=sym('t');

N0=1;
p0=zeros(1,n+1);
p0(N0)=1;
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(E,t,D);
% Example: p0*f(D,E,2) will compute the amount of p(2)
```

P11.70

L =

```
-1.0000   1.0000        0        0        0        0        0        0        0        0        0
 0.1000  -1.1000   1.0000        0        0        0        0        0        0        0        0
      0   0.2000  -1.2000   1.0000        0        0        0        0        0        0        0
      0        0   0.3000  -1.3000   1.0000        0        0        0        0        0        0
      0        0        0   0.4000  -1.4000   1.0000        0        0        0        0        0
      0        0        0        0   0.5000  -1.5000   1.0000        0        0        0        0
      0        0        0        0        0   0.6000  -1.6000   1.0000        0        0        0
      0        0        0        0        0        0   0.7000  -1.7000   1.0000        0        0
      0        0        0        0        0        0        0   0.8000  -1.8000   1.0000        0
      0        0        0        0        0        0        0        0   0.9000  -1.9000   1.0000
      0        0        0        0        0        0        0        0        0   1.0000  -1.0000
```

>> E

E =

```
-0.0061  -0.0865  -0.3386   0.6408   0.8545   0.9660   0.9937  -0.9508   0.3015   0.6446   0.8396
 0.0143   0.1544   0.4493  -0.5957  -0.4913  -0.2438   0.0436  -0.3017   0.3015   0.5157   0.4781
-0.0317  -0.2518  -0.5172   0.4301   0.1478  -0.0595  -0.0931  -0.0308   0.3015   0.3996   0.2362
 0.0657   0.3685   0.4929  -0.1947   0.0428   0.0519  -0.0314   0.0444   0.3015   0.2965   0.0861
-0.1259  -0.4721  -0.3509  -0.0065  -0.0561   0.0203   0.0171   0.0366   0.3015   0.2063   0.0040
 0.2207   0.5071   0.1280   0.0813  -0.0073  -0.0177   0.0202   0.0085   0.3015   0.1289  -0.0305
-0.3477  -0.4162   0.0696  -0.0317   0.0286  -0.0146   0.0024  -0.0114   0.3015   0.0645  -0.0347
 0.4799   0.1894  -0.1274  -0.0383   0.0051   0.0056  -0.0106  -0.0155   0.3015   0.0129  -0.0222
-0.5537   0.0856   0.0311   0.0310  -0.0194   0.0127  -0.0095  -0.0079   0.3015  -0.0258  -0.0039
 0.4803  -0.2359   0.0855   0.0267  -0.0085   0.0025   0.0004   0.0037   0.3015  -0.0516   0.0124
-0.2035   0.1321  -0.0645  -0.0287   0.0147  -0.0098   0.0090   0.0115   0.3015  -0.0645   0.0217
```

>> D

D =

```
-3.3609        0        0        0        0        0        0        0        0        0        0
      0  -2.7862        0        0        0        0        0        0        0        0        0
      0        0  -2.3267        0        0        0        0        0        0        0        0
      0        0        0  -1.9296        0        0        0        0        0        0        0
      0        0        0        0  -1.5749        0        0        0        0        0        0
      0        0        0        0        0  -1.2523        0        0        0        0        0
      0        0        0        0        0        0  -0.9561        0        0        0        0
      0        0        0        0        0        0        0  -0.6827        0        0        0
      0        0        0        0        0        0        0        0   0.0000        0        0
      0        0        0        0        0        0        0        0        0  -0.2000        0
      0        0        0        0        0        0        0        0        0        0  -0.4305
```

>> inv(E)

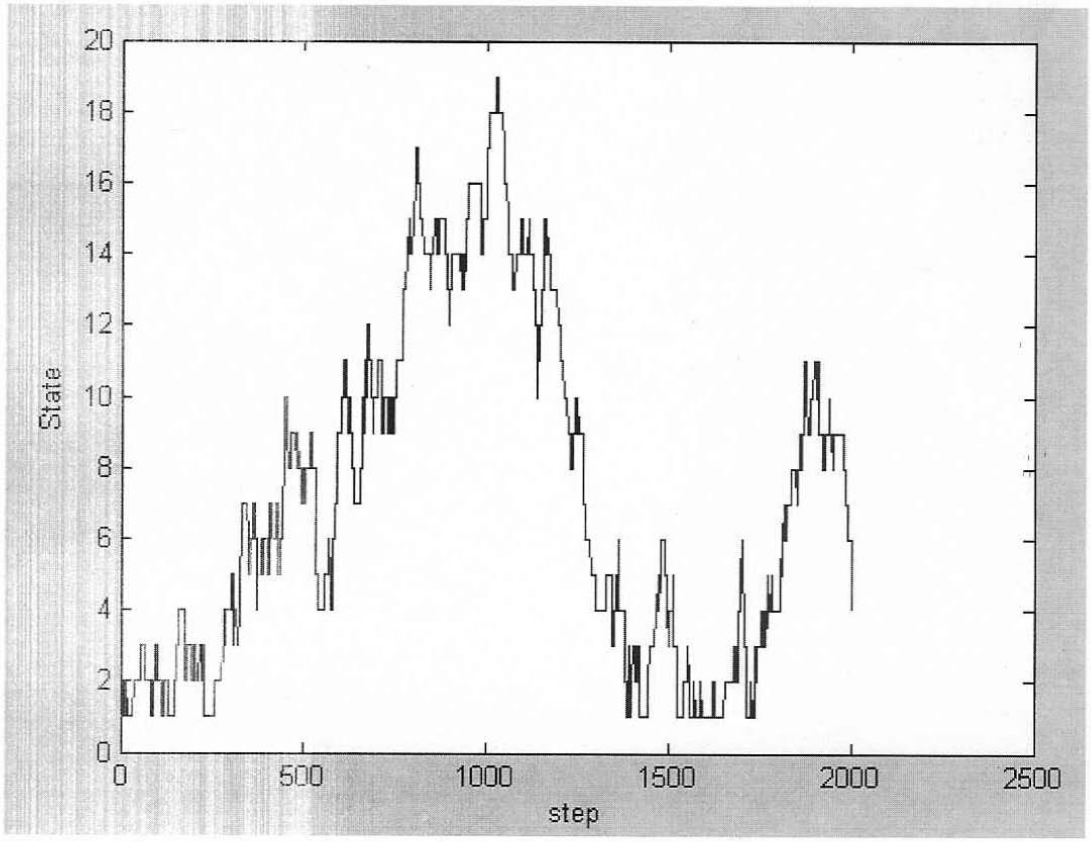
ans =

```
-0.0000   0.0001  -0.0007   0.0050  -0.0240   0.0842  -0.2212   0.4362  -0.6290   0.6064  -0.2568
-0.0001   0.0018  -0.0146   0.0710  -0.2275   0.4888  -0.6686   0.4347   0.2455  -0.7520   0.4210
-0.0017   0.0232  -0.1334   0.4237  -0.7540   0.5500   0.4988  -1.3036   0.3977   1.2157  -0.9164
 0.0178  -0.1656   0.5980  -0.9022  -0.0750   1.8838  -1.2237  -2.1150   2.1366   2.0423  -2.1970
 0.0970  -0.5574   0.8388   0.8100  -2.6536  -0.6919   4.5090   1.1502  -5.4563  -2.6428   4.5971
 0.2665  -0.6726  -0.8203   2.3850   2.3349  -4.0804  -5.5758   3.0600   8.6820   1.8831  -7.4624
 0.3625   0.1591  -1.6979  -1.9107   2.6021   6.1316   1.2214  -7.6360  -8.6272   0.3950   9.0000
-0.2363  -0.7498  -0.3828   1.8394   3.7956   1.7662  -3.9201  -7.6601  -4.8404   2.5022   7.8863
 0.0003   0.0026   0.0129   0.0430   0.1076   0.2152   0.3587   0.5124   0.6405   0.7117   0.7117
 0.0077   0.0615   0.2384   0.5896   1.0254   1.2817   1.0681   0.3052  -0.7629  -1.6954  -2.1193
 0.0687   0.3915   0.9668   1.1748   0.1368  -2.0844  -3.9434  -3.6106  -0.8003   2.7940   4.9060
```

11.71

```
Nmax=50;
P=zeros(Nmax+1,3);
mu=1;
lambda=.9;
delta=.1;
a=delta*lambda;  %lambda/(lambda+mu);
b=delta*mu;  %mu/(lambda+mu);
P(1,:)=[0,1-a,a];
r=[(1-a)*b,a*b+(1-a)*(1-b),(1-b)*a];
for n=2:Nmax;
   P(n,:)=r;
end
P(Nmax+1,:)=[(1-a)*b,1-(1-a)*b,0];
IC=zeros(Nmax+1,1);
IC(1,1)=1;
L=2000;

stseq=zeros(1,L);
s=[1:Nmax+1];
step=[-1,0,1];
Initst=1;
stseq(1)=Initst;
for n=2:L+1
   stseq(n)=stseq(n-1)+dscRnd(1,P(stseq(n-1),:),step);
end
```



Problems Requiring Cumulative Knowledge

11.72

8.30 a) If the process starts in state 0 at times 0, then the process will be in states {0,2} at even time instants and in states {1,3} at odd time instants. The transition probabilities between the states in {0,2} in 2 steps are

$$\mathbf{P}^2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$$

The transition probabilities after $2n$ steps are:

$$\mathbf{P}^{2n} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}^n = E \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix} E^{-1}$$

where $\mathbf{P}^2$ has eigenvalues $\lambda_1 = 1$, $\lambda_2 = -\frac{1}{2}$ and corresponding eigenvectors $\underline{e}_1 = [1, 1]$ and $\underline{e}_2 = [\frac{1}{3}, -\frac{2}{3}]$

$$\begin{aligned} \mathbf{P}^{2n} &= \begin{bmatrix} 1 & \frac{1}{3} \\ 1 & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \left(-\frac{1}{2}\right)^n \end{bmatrix} \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{3} + \frac{1}{3}\left(-\frac{1}{2}\right)^n & \frac{1}{3} - \frac{1}{3}\left(-\frac{1}{2}\right)^n \\ \frac{2}{3} - \frac{2}{3}\left(-\frac{1}{2}\right)^n & \frac{1}{3} + \frac{2}{3}\left(-\frac{1}{2}\right)^n \end{bmatrix} \\ &\to \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{aligned}$$

The transition probabilities for an odd number of steps are given by:

$$\begin{bmatrix} P_{01}(2n+1) & P_{03}(2n+1) \\ P_{21}(2n+1) & P_{23}(2n+1) \end{bmatrix} = \mathbf{P}^{2n} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{P}^{2n}$$
$$\to \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Thus at even time instants state 0 has probability $\frac{2}{3}$ and state 1 has probability $\frac{1}{3}$. At odd time instants state 1 has probability $\frac{2}{3}$ and state 2 has probability $\frac{1}{3}$. The stationary state pmf is then

$$\pi_0 = \pi_1 = \frac{1}{3} \qquad \pi_2 = \pi_3 = \frac{1}{6}$$

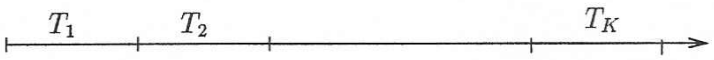
b) X_n is cyclostationary if its joint distributions are invariant with respect to time shifts that are multiples of some period M. If the above process is started with initial probabilities $[\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6}]$ then the joint distributions will be invariant with respect to even time shifts (i.e. $M = 2$), therefore the process is cyclostationary.

11.73

8.31 X_n is an ergodic Markov chain $\Rightarrow$ positive recurrent, aperiodic, irreducible

$$\text{time average of } I_j(n) = \overline{I} = \frac{\sum_T I_j(n)}{T}$$

Suppose the process has returned to state 'i' times as shown below:



Then

$$T = \sum_{i=1}^{K} T_j(i)$$

where $T_j(i)$ where $T_j(i)$ are interarrival times to state 'j'.

Therefore,

$$\overline{I} = \frac{K}{\sum_{i=1}^{K} T_j(i)}$$

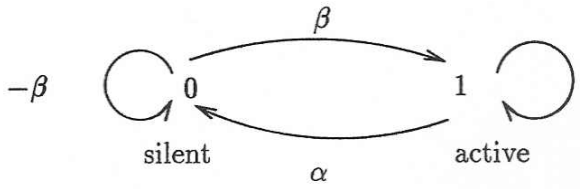
As

$$T \to \infty \Rightarrow \overline{I} \to \frac{1}{E[T_j]} = \pi_j$$

Therefore, the limiting value of time average of $I_j(n)$ is equal to long-term proportion of time spent in state 'j'. This result is an ergodic theorem.

11.74

.32 a) $X(t)$ has transition rate diagram:



b) The associated Chapman Kolmogorov equations are:

$$P_j'(t) = \sum_i \gamma_{ij} P_i(t)$$

$$\Rightarrow \begin{cases} P_0'(t) &= -\beta P_0(t) + \alpha P_1(t) \\ P_1'(t) &= t\alpha P_0(t) - \beta P_1(t) \end{cases}$$

$P_0(t)$ and $P_1(t)$ are obtained by solving this set of differential equations:

$$P_0(t) = P_0 + (P_0(0) - P_0)e^{-(\alpha+\beta)t}$$

$$P_1(t) = P_1 + (P_1(0) - P_1)e^{-(\alpha+\beta)t}$$

where $P_0 = \frac{\alpha}{\alpha+\beta}$ and $P_1 = \frac{\beta}{\alpha+\beta}$

c)
$$\begin{aligned} R_X(t_1,t_2) &= E[X(t_1)\cdot X(t_2)] \\ &= \sum_{i=0}^{1}\sum_{j=0}^{1} a_i a_j P[X(t_1)=a_i, X(t_2)=a_j] \end{aligned}$$

where a_i = state of the system = ϕ or 1

$$\begin{aligned} \Rightarrow R_X(t_1,t_2) &= P(X(t_1)=1, X(t_2)=1) \\ &= P(X(t_2)=1/X(t_1))\cdot P(X(t_1)=1) \end{aligned}$$

Without loss of generality, let's assume $t_2 \geq t_1$.
Let's suppose $P_0 = 1, P_1(0) = 0$ then

$$P(X(t_1)=1) = P_1\cdot(1-e^{-(\alpha+\beta)t_1})$$

$P(X(t_2)=1/X(t_1)=1)$ can be also obtained using the results of part (b) assuming that $\tau = t_2 - t_1$ and $P_0(0)=0$, $P_1(0)=1$:

$$\begin{aligned} P(X(t_2)=1/X(t_1)=1) &= P_1 + (1-P_1)e^{-(\alpha+\beta)(t_2-t_1)} \\ &= P_1 + P_0 e^{-(\alpha+\beta)\tau} \end{aligned}$$

$$\Rightarrow R_X(t_1,t_2) = P_1^2 + P_1P_0e^{-(\alpha+\beta)\tau} + P_1^2e^{-(\alpha+\beta)t_1} + P_1P_0e^{-(\alpha+\beta)(t_1+\tau)}$$

d) From the equation of $R_X(t_1,t_2)$, it is seen that $X(t)$ is not WSS. However, if $t_1 \to \infty$, the last two terms will die out and we have:

$$R_X(t_1,t_2) \to R_X(\tau) = P_1^2 + P_0P_1e^{-(\alpha+\beta)\tau}$$

and

$$\eta_X^{(t)} = E[X(T)] = P_1(T)\eta_X = P_1 \text{ as } t\to\infty$$

So, $X(t)$ is asymptotically WSS.
The power spectral density of $X(t)$ is:

$$S_X(f) = \mathcal{F}[R_X(\tau)]$$

$$\Rightarrow \quad S_X(W) = p_1^2\delta(f) + p_1p_0\frac{2(\alpha+\beta)}{(\alpha+\beta)^2+4\pi^2f^2}$$

e) If we have 'n' independent speakers with $X_i(t) = 0$ or 1 , $i = 1, ..., n$, then

$$\Rightarrow \quad N(t) = X_1(t) + X_2(t) + ... + X_n(t) = \sum_{i=1}^{n} X_i(t)$$

$$\begin{aligned} R_N(t_1, t_2) &= E[N(t_1) \cdot N(t_2)] = E\left[\sum_{i=1}^{n} X_i(t) \cdot \sum_{j=1}^{n} X_j(t)\right] \\ &= \sum_{i=1}^{n} E[X_i(t_1)X_i(t_2)] + \sum_{i=1}^{n}\sum_{j\neq i} E[X_i(t_1)X_j(t_2)] \\ &= n \cdot R_X(t_1, t_2) + \sum\sum E[X_i(t_1)] \cdot E[X_j(t_2)] \end{aligned}$$

$$R_N(t_1, t_2) = n \cdot R_X(t_1, t_2) + n(n-1)\eta_X(t_1) \cdot \eta_X(t_2)$$

Suppose $t_2 = t_1 + \tau$ and $\tau \geq 0$
If $t_1 \to \infty$

$$\Rightarrow R_N(t_1, t_2) = R_N(\tau) = n \cdot R_X(\tau) + n(n-1)\eta_X^2$$

$$\begin{aligned} S_N(f) &= \mathcal{F}[R_N(\tau)] = n \cdot S_X(f) + n(n-1)\eta_X^2 \cdot 2\delta(f) \\ S_N(f) &= n^2 p_1^2 \delta(f) + n p_1 p_o \frac{2(\alpha+\beta)}{(\alpha+\beta)^2 + 4\pi^2 f^2} \end{aligned}$$

11.75

8.33 X_n : continuous-valued discrete-time Markov process.

(a) The joint pdf of $(n+1)$ values of the process X_n is denoted by

$$f_{X_n \ldots X_0}(x_n, x_{n-1}, \ldots, x_1, x_0)$$

According to the theorem of conditional pdf we can write

$$f_{X_n,\ldots,X_0}(x_n, \ldots, x_0) = f_{X_n/X_{n-1},\ldots,X_0}(x_n/x_{n-1}, \ldots, x_0) \cdot f_{X_{n-1}\ldots X_0}(x_{n-1}, \ldots, x_0)$$

However, for a first-order Markov process, we have:

$$f_{X_n/X_{n-1},\ldots,X_0}(x_n/x_{n-1}, \ldots, x_0) = f_{X_n/X_{n-1}}(x_n/x_{n-1})$$

Continuing, this process for $f_{X_{n-1},\ldots,X_0}(x_{n-1}, \ldots, x_0)$, the following can be easily obtained:

$$f_{X_n,\ldots,X_0}(x_n, x_{n-1}, \ldots, x_0) = f_{X_n/X_{n-1}}(x_n/x_{n-1}) \cdot f_{X_{n-1}/X_{n-2}}(x_{n-1}/x_{n-2}), \ldots, f(x_0)$$

b) Consider the two-step transition pdf: $f_{X_{n+2}/X_n}(x_{n+2}/x_n)$

$$\begin{aligned} f_{X_{n+2}/X_n}(x_{n+2}/x_n) &= \frac{f_{X_{n+2}\cdot X_n}(x_{n+2}, x_n)}{f_{X_n}(x_n)} \\ &= \frac{\int f_{X_{n+2}X_{n+1}+X_n}(x_{n+2}, x_{n+1}.x_n)dx_{n+1}}{f_{X_n}(x_n)} \\ &= \int f_{X_{n+2}/X_{n+1}}(x_{n+2}/x_{n+1}) \cdot f_{X_{n+1}/X_n}(x_{n+1}/x_n) \cdot dx_{n+1} \\ \Rightarrow\ & f_{X_{n+2}/X_n}(x_{n+2}/x_n) = \int f_{X_{n+2}/X_{n+1}}(x_{n+2}/x_{n+1}) \\ & \cdot f_{X_{n+1}/X_n}(x_{n+1}/x_n)dx_{n+1} \end{aligned}$$